The sum of reciprocals of prime divisors of $\binom{2n}{n}$: the asymptotic for every n

Samuel Lavery

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Abstract

Erdős, Graham, Ruzsa and Straus (1975) conjectured that

$$\sum_{p \mid \binom{2n}{n}} \frac{1}{p} = (1 + o(1)) \log\log n.$$

We prove this for *every* sufficiently large n, in the explicit form $\log\log n + O\big((\log\log n)^{2/3}\big)$, executing the route sketched by Tao (comment on the Erdős problem registry, no. 377, 29 August 2025): a band-by-band analysis of the primes $p \in (n^{1/(k+1)}, n^{1/k}]$, in which non-divisibility forces the fractional parts $\{n/p^j\}$ into $[0, \frac{1}{2})$ at every level j, combined with equidistribution of these fractional parts over primes in each band. The analytic apparatus is developed here from first principles — Vaughan's identity, the van der Corput derivative tests with the order chosen adaptively, and a multidimensional Erdős–Turán inequality are all proved in Appendices A–C; the one imported theorem is the prime number theorem with a classical error term. The combinatorial core (the Kummer carry criterion, the band decomposition, and an exact carry-mass conservation law) is machine-verified in Lean 4.

1 Statement and architecture

Theorem 1 (Main)**.** *For every sufficiently large n,*

$$\sum_{\substack{p \le n \\ p \mid \binom{2n}{n}}} \frac{1}{p} = \log\log n + O\big((\log\log n)^{2/3}\big),$$

with an effective implied constant.

Since $\sum_{p \le n} 1/p = \log\log n + M + O(1/\log n)$, Theorem 1 is equivalent to the statement that the *non-divisor mass* $E(n) := \sum_{p \le n,\ p \nmid \binom{2n}{n}} 1/p$ satisfies $E(n) = O\big((\log\log n)^{2/3}\big)$ for every large n; this is what §8 proves. The full registry problem asks for $E(n) = O(1)$, which is a strictly stronger statement.

The proof has four components.

(K) **Carry criterion** (Kummer; machine-verified). For prime p, $p \nmid \binom{2n}{n}$ iff $\{n/p^j\} < \frac{1}{2}$ for every $j \ge 1$ with $p^j \le 2n$. [Lean: `prime_not_dvd_centralBinom_iff_carryFree`, file `Erdos377CarryRails.lean`, standard axiom footprint.]

(B) **Band decomposition** (machine-verified). Splitting primes by $k = \lfloor \log n/\log p \rfloor$, the band $(n^{1/(k+1)}, n^{1/k}]$ carries Mertens mass $\log(1+1/k) + O(\cdot) \asymp 1/k$, and within one band the level-one carry condition is a union of explicit "hyperbola" intervals. [Lean: `levelOneCarryPrimes_filter_eq_sdiff`, `levelOneCarryMass_eq_band_sum`.]

(E) **Band equidistribution** (the analytic core; §3). For $k \le K(n)$ and $J = J(k) \to \infty$ slowly, the vector $\left(\{n/p^{k-1}\}, \ldots, \{n/p^{k-J}\}\right)$ equidistributes as p ranges over the band, uniformly in n; hence the proportion of the band's mass carrying no carry at these J levels is $2^{-J}(1+o(1))$, and the carry-free proportion is at most that. The top level $j = k$ is deliberately not used (Proposition 4); its distribution is the hyperbola-weighted one of Lemma 2 and enters only as corroboration.

(A) **Assembly** (§8). Summing $2^{-J(k)}/k$ over $J(n) < k \le K(n)$ and treating the two flanking ranges $k \le J(n)$ and $k > K(n)$ trivially yields $E(n) = O\big((\log\log n)^{2/3}\big)$.

2 The exact density law

Within a band, the top-level condition is not uniformly distributed: the quotient $t = n/p^k$ is small near the band's top edge, and the correct density is the hyperbola-weighted one. This is an exact statement, not a model.

Since Proposition 4 excludes the top level, nothing in the chain from §3 to §8 consumes this section: it records what the top level does, and corroborates the architecture, but is not a hypothesis of the main theorem. It is used in one optional place only — Appendix B, §B.6(b), which the route of §B.6(d) bypasses.

Lemma 2 (Band measure)**.** *In the coordinate $t = n/p^k$ the Mertens measure $\sum 1/p$ of the band is log-uniform. Precisely, for $1 \le a < b \le n^{1/(k+1)}$,*

$$\sum_{\substack{p \in (n^{1/(k+1)},\, n^{1/k}] \\ a < n/p^k \le b}} \frac{1}{p} = \log \frac{\log n - \log a}{\log n - \log b} \;+\; O\Big(e^{-c\sqrt{\log n/(k+1)}}\Big)$$

$$= \frac{\log(b/a)}{\log n}\Big(1 + O\Big(\frac{\log b}{\log n}\Big)\Big) + O\Big(e^{-c\sqrt{\log n/(k+1)}}\Big),$$

with $c > 0$ and the implied constants absolute. Taking $a = 1$, $b = n^{1/(k+1)}$ recovers the band mass $\log(1+1/k)$ exactly, so the density is $1/\log n$ per unit of $\log t$, uniformly across the band.

Consequently, on the block $t \in (\nu, \nu+1]$ the carry strip $\{t\} \in [\frac{1}{2}, 1)$ carries the fraction

$$\sigma_\nu^{\mathrm{c}} = \frac{\log \frac{\nu+1}{\nu+1/2}}{\log \frac{\nu+1}{\nu}}\big(1 + o(1)\big),$$

which increases from $\log(4/3)/\log 2 = 0.41504\ldots$ at $\nu = 1$ to $\frac{1}{2}$ as $\nu \to \infty$: biased below $\frac{1}{2}$ at small quotients, tending to $\frac{1}{2}$ from below at large ones. The complementary carry-free fraction decreases from $\log(3/2)/\log 2 = 0.58496\ldots$ to $\frac{1}{2}$.

Proof. The condition $a < n/p^k \le b$ is $(n/b)^{1/k} \le p < (n/a)^{1/k}$, so by Mertens' theorem in the form $\sum_{p\le y} 1/p = \log\log y + M + O(e^{-c\sqrt{\log y}})$ — the prime number theorem with a classical error term — the sum equals $\log\log(n/a)^{1/k} - \log\log(n/b)^{1/k}$ up to the stated error, and the factor $1/k$ cancels between the two $\log\log$'s. Both evaluation points exceed $n^{1/(k+1)}$, which gives the error term. The second form is $\log \frac{1-\log a/\log n}{1-\log b/\log n}$ expanded. For $\sigma^{\mathrm{c}}_{\nu}$, apply the first form with $(a,b) = (\nu, \nu+\frac{1}{2})$ and $(\nu, \nu+1)$ and divide; the relative error is $o(1)$ as soon as $\nu \log n \le e^{\frac{c}{2}\sqrt{\log n/(k+1)}}$, which holds with an exponential margin for every ν that occurs in Appendix B (there $\nu \le T_0 = (\log n)^{O(1)}$). □

The finite band form is machine-verified as `levelOneCarryMass_eq_band_sum` and the two-level form as `levelTwoJointCarry_fiber_logMass`.

Remark 3 (Empirical corroboration; not an input to any proof)**.** *The following is measurement, recorded because the falsification it invites is cheap; no statement in this paper depends on it. For 42 values of $n \in [10^{11}, 10^{12}]$ the top-level counts match the hyperbola-weighted prediction of Lemma 2 with z-scores $-1.04, -1.02, -2.53$ at $k = 2, 3, 4$ (aggregated); each level below the top halves the count to within 10^{-3} relative. There is no anomalous "clock error": the only deviations from the naive uniform model are the deterministic hyperbola weights of Lemma 2.*

3 The equidistribution input

Fix a band index k and write $P = n^{1/k}$. The required statement is:

Proposition 4 (Band equidistribution)**.** *Let $k \le K(n)$ and*

$$J = J(k) := \min\big(\lceil(\log\log n)^{1/3}\rceil,\ k-1\big).$$

For any subintervals $I_1, \ldots, I_J \subseteq [0,1)$,

$$\#\{p \in (n^{1/(k+1)}, n^{1/k}] :\ \big\{n/p^{k-i}\big\} \in I_i\ (1 \le i \le J)\} = \Big(\prod_i |I_i| + o(1)\Big)\big(\pi(n^{1/k}) - \pi(n^{1/(k+1)})\big),$$

uniformly in n, with $o(1) \le (\log n)^{-1}$. The levels used are $j = k - i$ for $1 \le i \le J$, i.e. levels $k-J, \ldots, k-1$: the top level $j = k$ is deliberately excluded*, so that every frequency vector has amplitude at least Q on every block (§B.6(d)); the top level's hyperbola-weighted distribution (Lemma 2) is thereby corroborative, not load-bearing. Since the assembly requires only an upper bound on the carry-free count, discarding one level costs a factor 2 absorbed in the constants.*

Reduction to exponential sums. By the Erdős–Turán– Koksma inequality in J dimensions, Proposition 4 follows from

$$S(\mathbf{h}) \ = \sum_{p\in(n^{1/(k+1)},\, n^{1/k}]} e\Big(\sum_{i\le J} h_i \,\frac{n}{p^{k-i}}\Big) \ \ll\ \frac{\pi(P)}{(\log n)^B} \tag{1}$$

for every fixed B and all $0 < \|\mathbf{h}\|_\infty \le H$, with H a small power of log. (The exponents $k-i$, $1 \le i \le J$, are those of Proposition 4, the top level being excluded. The saving is measured against $\log n$, not $\log P$: the band carries $\gg P/\log n$ primes while $\pi(P) \asymp Pk/\log n$, so a saving in $\log P$ alone would be eaten by the factor $k \leq K(n)$ — see Appendix B, §B.9.)

Status of (1)**: the active core.** Partition the band into *fine blocks* $(Q, \rho Q]$ with $\rho = 1+1/\varrho$, $\varrho = k + r_{\max}$ (*not* dyadic blocks: on a dyadic block the two-sided ratio of the r-th derivative is 2^{j+r} with j as large as K, which destroys every derivative test; on a fine block the ratio is ≤ 5 absolutely, Lemma 21 in Appendix B). The phase for a frequency vector $\mathbf{h}$ is a sum of monomials; by Lemma 21 its top non-vanishing level i_0 dominates, and the governing amplitude is $F = |h_{i_0}|\, n\, Q^{-j}$, $j = k - i_0$, with $Q \leq F \leq HQ^{J+1}$ across the band. Vaughan's identity (Lemma 12, with $U = Q^{1/3}$) reduces (1) to two families:

(I) *Type I:* $$\sum_{m\sim M} \max_I \Big|\sum_{l\in I} e\big(h\, n/(ml)^j\big)\Big|, \qquad ML \asymp Q,\ M \leq Q^{2/3} \text{ (so } L \geq Q^{1/3}).$$

(II) *Type II (bilinear):* $$\sum_{m\sim M}\sum_{l\sim L} a_m b_l\, e\big(h\, n/(ml)^j\big), \qquad ML \asymp Q,\ Q^{1/3}/2 \leq M \leq Q^{1/2}$$ after the symmetry reduction of Lemma 28, $|a_m| \leq \tau(m)$, $|b_l| \leq \log l$.

Both families are proved here from first principles; no oscillatory-sum estimate is imported. The demand is a saving of a fixed power of $\log Q$ only (the assembly needs 2^{-J} with $J = J(k)$ as scheduled in Proposition 4), far below the power-of-Q savings pursued in the exponent-pair literature, and at this strength each tool reduces to its elementary core:

- Vaughan's identity: an elementary divisor-sum rearrangement (§5, self-contained, one page);
- the Type I bound: Weyl differencing plus the second-derivative test for the monomial phase $x \mapsto h\, n\, x^{-j}$ (§4, self-contained; in carrier language, a phase-cell cancellation statement);
- the Type II bound: Cauchy–Schwarz plus an elementary spacing count for the values $h\, n/(ml)^j$ (§6, self-contained).

The genuine work, and the reason for Tao's "lengthy but routine," is uniformity: the amplitude F varies from $\asymp Q$ to HQ^{J+1}, and every (F, Q) regime occurring for $k \leq K(n)$ must be covered. The derivative-test order is chosen *adaptively*, $r = \max(2, \lceil \log F/\log N\rceil + 1) \leq r_{\max} = 3J + 5$, the consecutive fixed-$r$ windows overlapping (Appendix B, §B.5); the $r = 2$ branch saves a power of the *amplitude alone*, so the cancellation range extends down to F a fixed power of $\log n$ (threshold T_0, (21)), and the residual regime $F < T_0$ is counted directly by band geometry with PNT in intervals of relative length $(\log n)^{-O(1)}$. The complete inputs of this paper are therefore: Kummer's carry criterion and the elementary central-binomial bounds (both machine-verified), the Chebyshev–Mertens bounds of §2 (machine-verified in finite form), the prime number theorem with a classical de la Vallée Poussin error term $\pi(y) = \mathrm{li}(y) + O(ye^{-c\sqrt{\log y}})$ (unconditional; consumed both for band masses at $k > \sqrt{\log n}$ and for the residual regime), and the elementary analysis of §§5–6 with Appendices A–C. The prime number theorem is the one item on that list not proved here; every other citation is provenance only.

Remark 5 (No shortcut through PNT alone)**.** *One might hope to avoid* (1) *for small k by applying the prime number theorem to each hyperbola component separately. This fails* *for a structural reason worth recording: by Lemma 2 the band's Mertens mass is log-uniform across quotient scales up to* $n^{1/(k+1)}$*, while PNT-with-classical-error resolves components only up to quotient* $\exp(c\sqrt{\log P})$ *— a vanishing fraction of the mass in every band. The bilinear estimates are therefore needed for* all *bands, not only deep ones; what varies with* k *is only the* (F, Q) *regime.*

Remark 6. *The schedule for* J *is fixed once, in Proposition 4:* $J(k) = \min(\lceil (\log\log n)^{1/3}\rceil, k-1)$*. Bands with* $k \le J(n)$ *contribute* $\sum_{k\le J(n)} 1/k \ll \log\log\log n$ *to the error, absorbed in the rate of Theorem 1; the assembly needs only* $J \to \infty$*, which is why the equidistribution demand* (1) *is mild.*

4 Elementary oscillatory-sum toolkit

All lemmas in this section are classical in substance; they are proved in full so that the paper is self-contained. Throughout, $e(x) = e^{2\pi i x}$ and $\|x\|$ is the distance from x to the nearest integer.

Lemma 7 (Kusmin–Landau)**.** *Let* $f \in C^1[a, b]$ *with* f' *monotone and* $\|f'(x)\| \ge \delta > 0$ *on* $[a, b]$*. Then* $\left|\sum_{a\le l\le b} e(f(l))\right| \le 2/(\pi\delta) + 1$*.*

Proof. Write $\theta_l = f(l + 1) - f(l) = f'(\xi_l)$ with $\xi_l \in (l, l + 1)$; by monotonicity the θ_l are monotone and $\|\theta_l\| \ge \delta$. With $g_l = \big(1 - e(\theta_l)\big)^{-1}$ we have $e(f(l)) = \big(e(f(l)) - e(f(l + 1))\big)g_l$, and Abel summation gives

$$\Big|\sum_l e(f(l))\Big| \le 2\max_l |g_l| + \sum_l |g_{l+1} - g_l|.$$

Since $|1 - e(\theta)| = 2|\sin \pi\theta| \ge 4\|\theta\|$, each $|g_l| \le (4\delta)^{-1}$. As $\theta \mapsto (1 - e(\theta))^{-1}$ has bounded variation $O(\delta^{-1})$ on each monotone stretch with $\|\theta\| \ge \delta$ (the image path is monotone along a circle arc staying at distance $\ge 4\delta$ from 0), the variation sum is $O(\delta^{-1})$. Tracking constants gives the stated bound. □

Lemma 8 (Second-derivative test)**.** *Let* $f \in C^2[a, b]$ *with* $\lambda \le f'' \le c\lambda$ *on* $[a, b]$*,* $\lambda > 0$*. Then* $\left|\sum_{a\le l\le b} e(f(l))\right| \ll_c (b - a)\lambda^{1/2} + \lambda^{-1/2}$*.*

Proof. f' is increasing with total increase $\le c\lambda(b-a)$. Fix $\delta \in (0, \frac{1}{4})$. The set where $\|f'\| < \delta$ is a union of at most $c\lambda(b - a) + 2$ intervals, each of length $\le 2\delta/\lambda$ (since f' moves at speed $\ge \lambda$); trivially estimate these ranges by their lengths. On the complementary intervals apply Lemma 7. This gives $|S| \ll (c\lambda(b-a)+1)\big(\delta/\lambda+\delta^{-1}\big)$; choosing $\delta = \lambda^{1/2}$ yields the claim. □

Lemma 9 (Weyl differencing)**.** *For any* $1 \le H \le N$ *and* $f : [1, N] \to \mathbb{R}$*,*

$$\Big|\sum_{l\le N} e(f(l))\Big|^2 \le \frac{2N}{H} \sum_{0\le h<H} \Big|\sum_{l\le N-h} e\big(f(l + h) - f(l)\big)\Big|.$$

Proof. Average the sum over H shifts, apply Cauchy–Schwarz, and expand the square; the diagonal terms give the h-sum. Standard bookkeeping. □

Lemma 10 (r-th derivative test)**.** *Let $r \geq 2$, let $\mathcal{I}$ be an interval containing $N \geq 1$ integers, and let $f \in C^r(\mathcal{I})$ satisfy $\lambda \leq |f^{(r)}| \leq A\lambda$ on $\mathcal{I}$, $A \geq 1$. Then*

$$\Big|\sum_{l\in\mathcal{I}} e(f(l))\Big| \leq C_0\, A^{2^{2-r}} \Big(N\lambda^{1/(2^r-2)} + N^{1-2^{2-r}}\lambda^{-1/(2^r-2)}\Big),$$

with C_0 an absolute constant, independent of r.

Proof. Appendix B, Lemma 19: the induction on r via Lemma 9 squares the sum before invoking the inductive hypothesis and takes a square root after, so the constants obey $C_r \leq (2+12C_{r-1})^{1/2}$ and converge to a fixed point < 13 rather than compounding. The choice of the Weyl parameter is exact, by the identity $\theta(1-2\mu_r) = 2\mu_r$. $\square$

Lemma 11 (Type I bound, adaptive order)**.** *Let $j \geq 1$ and let $g(l) = c\,n\,l^{-j}$ (amplitude $F \asymp |c|nL^{-j}$) on a fine block of N integers inside $l \sim L$. Note $g^{(r)}(l) = (-1)^r c\,n\,(j)_r\,l^{-j-r}$ with the Pochhammer factor $(j)_r = j(j+1)\cdots(j+r-1)$ (not j^r; for $j=1$ it is $r!$). Choose $r = \max\big(2, \lceil \log F/\log N\rceil + 1\big)$. Then, provided $F \geq T_0$ (a fixed power of $\log n$, (21)),*

$$\Big|\sum_{l} e(g(l))\Big| \ll N^{1-\eta'}, \qquad \eta' = \begin{cases} \frac{1}{4}\log F/\log N, & r=2 \quad (F \leq N),\\ \geq 2^{-r_{\max}-1}, & r \geq 3,\end{cases}$$

uniformly over all regimes arising in this paper ($r \leq r_{\max} = 3J+5$). The consecutive fixed-r windows overlap, so the union covers every amplitude $T_0 \leq F \leq HQ^{J+1}$.

Proof. Appendix B, Lemma 23 (which carries $(j)_r$ explicitly and verifies the window overlap, Remark 24). At $r=2$ the saving is a power of the amplitude alone, which is what extends the range down to $F \asymp (\log n)^{O(1)}$. $\square$

5 Vaughan decomposition

Lemma 12 (Vaughan)**.** *Let $U \geq 2$ and $Q \geq U^2$. For any function $\phi : (Q, 2Q] \to \mathbb{C}$ with $|\phi| \leq 1$,*

$$\Big|\sum_{p\sim Q}\phi(p)\Big| \ll \frac{Q}{U} + (\log Q)^2 \max_{M\leq U^2} \sum_{m\sim M} \max_{I} \Big|\sum_{l\in I:\ ml\sim Q} \phi(ml)\Big| + (\log Q)^2 \max_{U/2\leq M\leq Q/U} \big|\mathcal{B}_M(\phi)\big|,$$

where $\mathcal{B}_M(\phi) = \sum_{m\sim M}\sum_{l:\ ml\sim Q} a_m b_l\,\phi(ml)$ for certain coefficients with $|a_m| \leq \tau(m)$, $|b_l| \leq \log l$.

Proof. Apply the identity $\Lambda = \mu_{\leq U} * \log - \mu_{\leq U} * \Lambda_{\leq U} * 1 + \mu_{>U} * \Lambda_{>U} * 1$ to $\sum_{q\sim Q}\Lambda(q)\phi(q)$, split all convolution ranges dyadically, and pass from Λ-weights to primes by partial summation (the prime powers $q = p^i$, $i \geq 2$, contribute $O(Q^{1/2}\log Q)$). The first two convolutions produce the Type I term (the inner variable carries the smooth weight log or 1, removable by partial summation); the third produces the bilinear term with the stated coefficient bounds. The computation is carried out in full in Appendix A; see Lemma 16 there for the precise form proved. The amendments (i) and (ii) of Remark 17 — the absolute value inside the m-sum, and the endpoint $U/2$ — are already incorporated in the statement above; (iii), that in the bilinear term the inner variable likewise runs over a subinterval, the cut being on the product ml, is a convention carried through Appendix B. $\square$

6 The bilinear bound

Lemma 13 (Type II bound, effective form)**.** *Let $j \ge 1$, $\Phi(m,l) = h\,n/(ml)^j$ with governing amplitude $F = |h|\,n/Q^j$ on $ml \asymp Q$, and let $Q^{1/3}/2 \le M \le Q^{1/2}$, $L = Q/M$, with coefficients $|a_m| \le \tau(m)$, $|b_l| \le \log l$. If $Fj \ge e^{80}\varrho^{10}$, then*

$$|\mathcal{B}_M| \ \ll\ Q\,(\log Q)^{5/2}\Big(L^{-1/2} + (Fj)^{-1/10} + M^{-2^{-r_{\max}-2}}\Big).$$

Proof. Appendix B, Lemma 26. The structure: split the l-range into $\ll \varrho$ fine blocks, Cauchy–Schwarz in m (absorbing $\tau(m)^2$ by Lemma 15); the diagonal gives the $L^{-1/2}$; for $l \ne l'$ the differenced phase in m is a monomial of amplitude $G \asymp F\,j\,|l-l'|/L$ (the factor j from the mean value theorem); pairs with G below the threshold $G_0 = (Fj)^{4/5}$ are *counted* rather than estimated, and the remainder is estimated by Lemma 11 in m. The exponent $4/5$ balances the two contributions. The exchange of the roles of m and l needed to reach $M \le Q^{1/2}$ is Lemma 28, which uses only $\sum |a|^2 \ll M(\log 2M)^3$ and $|b|_\infty \le \log Q$. □

7 Scheduling: covering every regime

The parameters are chosen diagonally in n:

$$\varepsilon_n = (\log\log n)^{-1/3}, \qquad J(n) = \lceil (\log\log n)^{1/3}\rceil, \qquad K(n) = (\log n)^{1-\varepsilon_n}, \qquad H(n) = (\log n)^2.$$

For a band $k \le K(n)$, a fine block $(Q, \rho Q]$ inside it, and a frequency vector with top non-vanishing level $i_0 \le J(k)$, the governing amplitude is $F = |h_{i_0}|\,n\,Q^{-(k-i_0)}$ with $1 \le |h_{i_0}| \le H$; by the exclusion of the top level in Proposition 4, $F \ge Q$ always, and $F \le H\,Q^{\,J+1}$. With the adaptive order $r = \max(2, \lceil \log F/\log N\rceil + 1) \le r_{\max} = 3J + 5$ in Lemmas 11 and 13, the saving exponent is

$$\eta' \gg 2^{-r_{\max}} \ge \exp\big(-c(\log\log n)^{1/3}\big),$$

so that

$$L^{\eta'} \ge \exp\Big(c'\,(\log n)^{\varepsilon_n}\, e^{-c(\log\log n)^{1/3}}\Big) \ \gg\ (\log n)^A$$

for every fixed A, uniformly over $k \le K(n)$: the block length satisfies $\log L \asymp \log Q \ge \frac{1}{3}\log Q \gg (\log n)^{\varepsilon_n} = \exp\big((\log\log n)^{2/3}\big)$, which dominates $2^{r_{\max}} = \exp\big(O((\log\log n)^{1/3})\big)$. *The full regime verification — that every (F, Q, i, k) triple arising in the theorem falls into the derivative-test range of Lemma 11 (possible since $F \ge Q \ge T_0$ throughout after the top-level exclusion), with the residual near-stationary regime arising only in the corroborative top-level analysis — is set out case-by-case in Appendix B.*

Combining Lemmas 12–13 with the J-dimensional Erdős–Turán inequality (proved with product Fejér kernels in Appendix C; the weak form with (log)-power losses suffices) yields Proposition 4 with $o(1) \le (\log n)^{-1}$, uniformly in n and $k \le K(n)$.

8 Assembly

Assuming Proposition 4:

$$E(n) \ \le\ \sum_{k \le K}\Big(2^{-J(k)} + o(1)\Big)\Big(\log\big(1 + \tfrac{1}{k}\big) + O(\cdot)\Big) \ +\ \sum_{K < k \le \log_2 n} \frac{C}{k}$$

with $K = K(n) = (\log n)^{1-\varepsilon_n}$. With the schedule of Proposition 4 the first sum is $\ll 2^{-J(n)} \log K + \log\log\log n = e^{-c(\log\log n)^{1/3}} \log\log n + O(\log\log\log n)$ (the $\log\log\log n$ from the bands $k \le J(n)$, treated trivially); the second is $\le C \log \frac{\log_2 n}{K} = C\varepsilon_n \log\log n + O(1) = C(\log\log n)^{2/3} + O(1)$. Both are $o(\log\log n)$, and the second dominates, giving the explicit rate

$$\sum_{\substack{p \le n \\ p \mid \binom{2n}{n}}} \frac{1}{p} = \log\log n + O\big((\log\log n)^{2/3}\big),$$

with effective constants throughout. The band-Mertens error $O(\cdot)$ uses the prime number theorem with classical error term for $k > \sqrt{\log n}$ (Appendix B, §B.9); its finite band form is machine-verified as `primeHarmonicMass_railBand_le_uniform`.

Provenance and verification

The route is Tao's sketch (Erdős problem registry no. 377, comment of 29 Aug 2025); the contribution here is its execution.

Inputs. The proof consumes exactly four things.

(i) Kummer's carry criterion and the elementary central-binomial bounds — component (K), machine-verified.

(ii) The band decomposition and the carry-mass conservation law — component (B), machine-verified.

(iii) The prime number theorem with a classical de la Vallée Poussin error term, $\pi(y) = \mathrm{li}(y) + O(ye^{-c\sqrt{\log y}})$. It is used for the band Mertens masses at $k > \sqrt{\log n}$, where the elementary Mertens error $O(k/\log n)$ exceeds the main term $\asymp 1/k$, and — on the route of Appendix B, §B.6(b) — for the residual regime. This is the one theorem imported rather than proved here; it is unconditional and effective.

(iv) The analysis of §§4–6 and Appendices A–C, all proved in full: Vaughan's identity, the Kusmin–Landau and van der Corput tests with the order chosen adaptively, the bilinear estimate, and a J-dimensional Erdős–Turán inequality.

Lean anchors. Components (K) and (B), the conservation law $\sum_p \big(\text{carries}\big) \log p = \log \binom{2n}{n}$, and the reduction architecture are formalized in Lean 4 over Mathlib, in the file

`RequestProject/Erdos377CarryRails.lean`

($\sim$7,600 lines, axiom footprint {`propext`, `Classical.choice`, `Quot.sound`}, no `sorry`). The anchors are:

(K) `prime_not_dvd_centralBinom_iff_carryFree`;

(B) `levelOneCarryPrimes_filter_eq_sdiff`,
`levelOneCarryMass_eq_band_sum`,
`levelTwoJointCarry_fiber_logMass`
(the last two also give the finite form of Lemma 2);

(B′) `primeHarmonicMass_railBand_le_uniform`, the band-Mertens bound used in §8.

Component (E) — §3 together with Appendices A–C — is classical analytic number theory and is not formalized.

A The Vaughan computation

Throughout this appendix $U \geq 2$ and $Q \geq U^2$ are real parameters and $\phi : (Q, 2Q] \to \mathbb{C}$ satisfies $|\phi| \leq 1$. We write $*$ for Dirichlet convolution, $\mathbf{1}$ for the constant arithmetic function 1, δ for the convolution identity ($\delta(1) = 1$, $\delta(q) = 0$ for $q > 1$), and, for a real $V > 0$,

$$\mu_{\leq V} = \mu \cdot \mathbf{1}_{[1,V]}, \qquad \mu_{>V} = \mu - \mu_{\leq V}, \qquad \Lambda_{\leq V} = \Lambda \cdot \mathbf{1}_{[1,V]}, \qquad \Lambda_{>V} = \Lambda - \Lambda_{\leq V}.$$

The only facts about arithmetic functions used are $\mu * \mathbf{1} = \delta$, $\Lambda * \mathbf{1} = \log$ and hence $\mu * \log = \Lambda$; all three are Möbius inversion.

A.1 The identity

Lemma 14 (Vaughan's identity)**.** *For every integer $q > U$,*

$$\Lambda(q) = \big(\mu_{\leq U} * \log\big)(q) - \big(\mu_{\leq U} * \Lambda_{\leq U} * \mathbf{1}\big)(q) + \big(\mu_{>U} * \Lambda_{>U} * \mathbf{1}\big)(q).$$

Proof. Expand the last convolution using $\mu_{>U} = \mu - \mu_{\leq U}$ and $\Lambda_{>U} = \Lambda - \Lambda_{\leq U}$. Convolution is commutative and bilinear, so

$$\begin{aligned}
\mu_{>U} * \Lambda_{>U} * \mathbf{1} &= \mu * \Lambda * \mathbf{1} - \mu * \Lambda_{\leq U} * \mathbf{1} - \mu_{\leq U} * \Lambda * \mathbf{1} + \mu_{\leq U} * \Lambda_{\leq U} * \mathbf{1} \\
&= \mu * \log - \Lambda_{\leq U} * (\mu * \mathbf{1}) - \mu_{\leq U} * \log + \mu_{\leq U} * \Lambda_{\leq U} * \mathbf{1} \\
&= \Lambda - \Lambda_{\leq U} - \mu_{\leq U} * \log + \mu_{\leq U} * \Lambda_{\leq U} * \mathbf{1},
\end{aligned}$$

where the second line used $\Lambda * \mathbf{1} = \log$ twice and the third used $\mu * \mathbf{1} = \delta$ and $\mu * \log = \Lambda$. Rearranging,

$$\Lambda = \mu_{\leq U} * \log - \mu_{\leq U} * \Lambda_{\leq U} * \mathbf{1} + \mu_{>U} * \Lambda_{>U} * \mathbf{1} + \Lambda_{\leq U},$$

an identity of arithmetic functions. The final term vanishes at every $q > U$. $\square$

A.2 The divisor moment

The Type II proof (Lemma 13) consumes the following mean value; we prove it from Mertens' theorem only.

Lemma 15. *For $x \geq 2$, $\sum_{m \leq x} \tau(m)^2 \ll x(\log x)^3$. Consequently $\sum_{m \sim M} \tau(m)^2 \ll M(\log 2M)^3$ for $M \geq 1$.*

Proof. Define $h := \tau^2 * \mu$, so that $\tau^2 = h * \mathbf{1}$; as a convolution of multiplicative functions h is multiplicative, and

$$h(p^a) = \tau(p^a)^2 - \tau(p^{a-1})^2 = (a+1)^2 - a^2 = 2a + 1 > 0.$$

Hence $h \geq 0$ and, exchanging the order of summation,

$$\sum_{m \leq x} \tau(m)^2 = \sum_{d \leq x} h(d) \lfloor x/d \rfloor \leq x \sum_{d \leq x} \frac{h(d)}{d} \leq x \prod_{p \leq x} \Big(1 + \sum_{a \geq 1} \frac{2a+1}{p^a}\Big),$$

the last step because every $d \leq x$ has all its prime factors $\leq x$ and all terms are non-negative. For $0 < x_0 \leq \frac{1}{2}$,

$$\sum_{a \geq 1} (2a+1) x_0^a = \frac{2x_0}{(1-x_0)^2} + \frac{x_0}{1-x_0} = \frac{x_0(3-x_0)}{(1-x_0)^2},$$

and multiplying out,

$$(1-x_0)^3 + x_0(3-x_0)(1-x_0) = (1 - 3x_0 + 3x_0^2 - x_0^3) + (3x_0 - 4x_0^2 + x_0^3) = 1 - x_0^2 \leq 1,$$

so $1 + x_0(3-x_0)(1-x_0)^{-2} \leq (1-x_0)^{-3}$. Taking $x_0 = 1/p \leq \frac{1}{2}$ gives

$$\sum_{m \leq x} \tau(m)^2 \leq x \prod_{p \leq x} \Big(1 - \frac{1}{p}\Big)^{-3}.$$

Finally $\log \prod_{p \leq x} (1 - 1/p)^{-1} = \sum_{p \leq x} \big(1/p + O(1/p^2)\big) = \log\log x + O(1)$ by Mertens' theorem $\sum_{p \leq x} 1/p = \log\log x + M + O(1/\log x)$ (elementary; the finite form used elsewhere in this paper is machine-verified), whence $\prod_{p \leq x} (1 - 1/p)^{-3} \ll (\log x)^3$. The dyadic form follows by $\sum_{m \sim M} \tau(m)^2 \leq \sum_{m \leq 2M} \tau(m)^2$, the case $M < 2$ being trivial. $\square$

A.3 The three sums

For $Q < T \leq 2Q$ set

$$S(T) := \sum_{Q < q \leq T} \Lambda(q) \phi(q).$$

Every q occurring satisfies $q > Q \geq U^2 \geq U$, so Lemma 14 applies termwise and $S(T) = \Sigma_1(T) - \Sigma_2(T) + \Sigma_3(T)$ with

$$\Sigma_1(T) = \sum_{d \leq U} \mu(d) \sum_{l:\ Q < dl \leq T} \phi(dl) \log l, \tag{2}$$

$$\Sigma_2(T) = \sum_{m} c_m \sum_{l:\ Q < ml \leq T} \phi(ml), \qquad c_m := \sum_{\substack{de = m \\ d \leq U,\ e \leq U}} \mu(d) \Lambda(e), \tag{3}$$

$$\Sigma_3(T) = \sum_{Q < ml \leq T} w(m)\, \Lambda_{>U}(l)\, \phi(ml), \qquad w(m) := \sum_{d \mid m,\ d > U} \mu(d). \tag{4}$$

Here (2) is the convolution $\mu_{\leq U} * \log$ written out with $q = dl$ (so that the inner variable carries the weight $\log l$); (3) collects the two inner variables of $\mu_{\leq U} * \Lambda_{\leq U} * \mathbf{1}$ into $m = de$, the third variable becoming l; and (4) writes $\mu_{>U} * \Lambda_{>U} * \mathbf{1}$ with m carrying $\mu_{>U} * \mathbf{1}$ and l carrying $\Lambda_{>U}$.

Three elementary properties of the coefficients:

- c_m is supported on $m \leq U^2$ and $|c_m| \leq \sum_{de=m} \Lambda(e) = \log m \leq 2 \log U \leq \log Q$.
- $|w(m)| \leq \sum_{d \mid m} 1 = \tau(m)$, and w is supported on $m > U$: for $m \leq U$ every divisor of m is $\leq U$, so $w(m) = \delta(m) - \sum_{d \mid m} \mu(d) = \delta(m) - \delta(m) = 0$.
- $\Lambda_{>U}(l)$ is supported on $l > U$ and $0 \leq \Lambda_{>U}(l) \leq \log l$.

A.4 Removing the smooth weight from Σ_1

Only Σ_1 carries a weight depending on the inner variable. Fix $d \le U$ and put $A(u) := \sum_{Q/d<l\le u} \phi(dl)$ for $u \in [Q/d, T/d]$. Partial summation (Riemann–Stieltjes, or the finite Abel identity, applied to log which is C^1) gives

$$\sum_{Q/d<l\le T/d} \phi(dl)\log l = A(T/d)\log(T/d) - \int_{Q/d}^{T/d} A(u)\,\frac{du}{u},$$

whence, since $\int_{Q/d}^{T/d} du/u = \log(T/Q) \le \log 2$,

$$\Big|\sum_{Q/d<l\le T/d} \phi(dl)\log l\Big| \le \big(\log(2Q)+\log 2\big)\sup_u |A(u)| \le 2\log(2Q)\max_I\Big|\sum_{l\in I}\phi(dl)\Big|, \tag{5}$$

the maximum being over subintervals $I \subseteq (Q/d, T/d]$. (Only A itself, not $|A|$, occurs, so no divisor-type loss is incurred.)

It is this appearance of a maximum over subintervals — forced by the removal of $\log l$, and independently by the cutoff at T — that makes the Type I object below an inner sum over an *interval* rather than over a full dyadic block; the van der Corput estimates of §4 are stated on an arbitrary $[a, b]$ and are insensitive to this.

A.5 Dyadic splitting

Define, for $M \ge 1$ and $Q < T \le 2Q$,

$$\mathcal{T}_M(T) := \sum_{m\sim M}\ \max_{I\subseteq(Q/m,T/m]}\Big|\sum_{l\in I}\phi(ml)\Big|,$$

$$\mathcal{B}_M(T) := \max_I\Big|\sum_{m\sim M}\sum_{l\in I_m} a_m b_l\,\phi(ml)\Big|,$$

where in $\mathcal{B}_M$ the coefficients satisfy $|a_m| \le \tau(m)$, $|b_l| \le \log l$, and the inner range is $I_m = \{l : Q < ml \le T'\}$ for a single $T' \in (Q, T]$ (i.e. the constraint is a cut on the product ml, uniform in m).

Σ_1. Split $d \le U$ into the $\le \log_2 U + 1$ dyadic ranges $d \sim D$, $D = U2^{-i}$, $0 \le i \le \log_2 U$. By $|\mu| \le 1$ and (5),

$$|\Sigma_1(T)| \le 2\log(2Q)\,(\log_2 U+1)\max_{D\le U}\mathcal{T}_D(T) \ll (\log Q)^2 \max_{M\le U^2}\mathcal{T}_M(T). \tag{6}$$

Σ_2. Split $m \le U^2$ dyadically into $\le \log_2(U^2) + 1$ ranges and use $|c_m| \le \log Q$:

$$|\Sigma_2(T)| \le \log Q\,\big(2\log_2 U+1\big)\max_{M\le U^2}\mathcal{T}_M(T) \ll (\log Q)^2\max_{M\le U^2}\mathcal{T}_M(T). \tag{7}$$

Σ_3. Here $m > U$ and $l > U$; since $ml \le T \le 2Q$ we get $m < 2Q/U$. Cover $(U, 2Q/U]$ by the blocks $(M_i, 2M_i]$ with $M_i = (Q/U)2^{-i}$, $0 \le i \le I := \lceil\log_2(Q/U^2)\rceil$; the top block reaches $2Q/U$ and the bottom one has $M_I \in [U/2, U]$, so all M_i lie in $[U/2,\ Q/U]$ and $I + 1 \ll \log Q$. With $a_m := w(m)$ ($|a_m| \le \tau(m)$) and $b_l := \Lambda_{>U}(l)$ ($|b_l| \le \log l$),

$$|\Sigma_3(T)| \ll \log Q \max_{U/2\le M\le Q/U}\mathcal{B}_M(T). \tag{8}$$

A.6 Prime powers and passage to primes

Let $W(T) := \sum_{Q<p\le T} \phi(p)\log p$. Then $S(T) - W(T) = \sum_{Q<p^i\le T,\, i\ge 2} \phi(p^i)\log p$, and

$$\Big|\sum_{\substack{Q<p^i\le 2Q\\ i\ge 2}} \phi(p^i)\log p\Big| \le \sum_{2\le i\le \log_2(2Q)} \sum_{p\le (2Q)^{1/i}} \log p \ll Q^{1/2} + Q^{1/3}\log Q \ll Q^{1/2}\log Q,$$

using only Chebyshev's bound $\sum_{p\le y}\log p \ll y$ (elementary) and the fact that there are $\ll \log Q$ values of i. Hence

$$\sup_{Q<T\le 2Q} |W(T)| \le \sup_{Q<T\le 2Q} |S(T)| + O\big(Q^{1/2}\log Q\big). \tag{9}$$

Removing the weight $\log p$ by partial summation,

$$\sum_{p\sim Q} \phi(p) = \frac{W(2Q)}{\log 2Q} + \int_Q^{2Q} \frac{W(T)}{T\log^2 T}\,dT,$$

so that, since $\int_Q^{2Q} dT/(T\log^2 T) \le \log 2/\log^2 Q$,

$$\Big|\sum_{p\sim Q}\phi(p)\Big| \le \frac{2}{\log Q} \sup_{Q<T\le 2Q} |W(T)|. \tag{10}$$

A.7 Conclusion

Lemma 16 (Vaughan, appendix form)**.** *Let $U \ge 2$, $Q \ge U^2$, and $\phi : (Q, 2Q] \to \mathbb{C}$ with $|\phi| \le 1$. Then*

$$\Big|\sum_{p\sim Q}\phi(p)\Big| \ll \frac{Q}{U} + (\log Q)^2 \max_{M\le U^2} \sup_{Q<T\le 2Q} \mathcal{T}_M(T) + (\log Q)^2 \max_{U/2\le M\le Q/U} \sup_{Q<T\le 2Q} \mathcal{B}_M(T),$$

with $\mathcal{T}_M, \mathcal{B}_M$ as in §A.5 and coefficient bounds $|a_m| \le \tau(m)$, $|b_l| \le \log l$.

Proof. Combine (6), (7), (8) into $|S(T)| \ll (\log Q)^2 \max_{M\le U^2} \mathcal{T}_M(T) + (\log Q)\max_M \mathcal{B}_M(T)$, insert into (9) and (10), and divide by $\log Q$ (which only helps). The prime-power term contributes $O(Q^{1/2})$, and $Q^{1/2} \le Q/U$ because $U \le Q^{1/2}$. □

Remark 17 (Amendments to Lemma 12)**.** *Two features of Lemma 16 differ from the statement of Lemma 12 in the main text, and Lemma 12 should be read as amended.*

(i) The Type I term carries the absolute value inside the m-sum. *Lemma 12 displays $\big|\sum_{m\sim M}\sum_l \phi(ml)\big|$; the computation above delivers $\sum_{m\sim M}\big|\sum_l \phi(ml)\big|$. The two differ essentially: the coefficients $\mu(d)$ in (2) and c_m in (3) depend on m and are not of constant sign, so they cannot be removed from inside the m-sum by partial summation, only pulled out in absolute value at the cost of the factor* $\log Q$ *already present. The amended form is the one the paper uses: item (I) of §3 already writes the Type I family as $\sum_{m\sim M}\big|\sum_{l\sim L} e(\cdot)\big|$.*

(ii) Ranges. *The Type I range $M \le U^2$ is exactly as stated. The bilinear range is $U/2 \le M \le Q/U$ rather than $U \le M \le Q/U$: with $m \sim M$ meaning $M < m \le 2M$, a dyadic ladder inside $(U, 2Q/U]$ cannot have all its base points $\ge U$ and simultaneously reach $2Q/U$. The half is harmless — Lemma 13 is applied in Appendix B only for M exceeding a fixed power of* $\log Q$.

(iii) Inner range. *In both terms the inner variable runs over a subinterval, not a full dyadic block; see §A.4.*

B Regime verification

This appendix is the audit pass announced in §7. It does four things: it sharpens Lemma 10 to the form the application actually needs and tracks the constant in r (§B.1–B.2); it replaces the dyadic block by the block scale on which the implied constants are absolute (§B.3); it re-derives the Type I and Type II inputs on that scale, with the choice of r made adaptively (§B.4–B.5); and it enumerates every (F, Q, i, k) regime, verifying coverage (§B.6–B.8). Ten statements of the main text required amendment; each is flagged [**A**n] at the point where it arises and collected in §B.9.

Remark 18 (Reading this appendix against the amended main text)**.** *The notation of the audit is retained verbatim, because the flags [**A1**]–[**A10**] are the record of what had to change. Two conventions must be aligned when reading it beside §§3–7 as they now stand.*

Level index. *Here levels are indexed by i with exponent $j_i = k-i+1$, so $i = 1$ is the top level $j = k$. The amended Proposition 4 indexes the same levels by i' with exponent $k - i'$, that is $i' = i-1$, and runs $i' = 1, \ldots, J(k)$. Its level set is therefore $i = 2, \ldots, J(k)+1$ in the notation of this appendix — exactly the route of §B.6(d). Accordingly* (12) *sets $J(k) = \min\{J, k-1\}$, which is what makes $j_i \geq 1$ hold on that level set, and* (16) *and* (13) *are stated for $i_0 \leq J+1$: $F \leq HQ^{J+1}$ and $r_{\max} = 3J + 5$.*

What that excludes. *Every case with $i_0 = 1$ — rows 2 and 3 of the table in §B.7, the threshold T_0 of* (21)*, and all of §B.6(a)–(c) — then lies outside the main line, and serves only the corroborative top-level analysis of §2. Row 1 alone carries the theorem.*

Flag status. *All ten flags are discharged in the main text as it now stands: [**A1**] by the schedule in Proposition 4; [**A2**] by the amplitude convention of §3; [**A3**] by Lemmas 10 and 11; [**A4**] by the fine-block partition of §3; [**A5**] by the Pochhammer factor in Lemma 11; [**A6**] by Lemma 13 and the adaptive order of §7; [**A7**] by the top-level exclusion; [**A8**], [**A8**$'$] and [**A9**] by §7; [**A10**] by the input list of §3. Two further items raised by the re-indexing — the amplitude range with its knock-on to $r_{\max}$, and the exponent against which the target of* (1) *is measured — are recorded as [**B1**] and [**B2**] at the end of §B.9 and are likewise resolved in the text.*

B.0 Notation and the parameters

Throughout, n is large, and

$$\varepsilon_n = (\log\log n)^{-1/3}, \quad J = \lceil (\log\log n)^{1/3} \rceil, \quad K = (\log n)^{1-\varepsilon_n}, \quad H = (\log n)^2,$$

as in §7. Fix a band index $k \leq K$ and put

$$P := n^{1/k}, \qquad P_0 := n^{1/(k+1)}, \qquad \text{band} = (P_0, P].$$

For p in the band write $t = t(p) := n/p^k$; then t decreases from P_0 (at $p = P_0$, since $n/P_0^k = P_0^{k+1-k} = P_0$) to 1 (at $p = P$), so

$$1 \leq t \leq P_0 \leq Q \qquad \text{for every } p \text{ in the band.} \tag{11}$$

Levels are indexed by i with exponent $j_i := k - i + 1$; the level-i coordinate is $\{n/p^{j_i}\}$. We set

$$J(k) := \min\{J, k-1\}, \tag{12}$$

so that $j_i \geq 1$ for every $i \leq J(k)+1$ — the level set $i = 2, \ldots, J(k)+1$ being the operative one, by Remark 18. **[A1]** Proposition 4 and §7 must be read with $J(k)$ in place of J: for $k < J$ the band has only k levels and the coordinate $\{n/p^{k-i+1}\}$ is undefined for $i > k$. (This costs nothing: in §8 the terms $k < J$ contribute $\sum_{k<J} 2^{-k}/k = O(1)$.) The hypothesis "$J \leq c \log k$" of Proposition 4 is likewise inconsistent with the schedule $J = \lceil (\log\log n)^{1/3} \rceil$ of §7 and with $J(k) = \lfloor c \log k \rfloor$ of the remark following Remark 5; see **[A6]**.

Finally set

$$r_{\max} := 3J+5, \qquad \varrho := k + r_{\max} = k + 3J + 5, \qquad \rho := 1 + \frac{1}{\varrho}. \tag{13}$$

For a frequency vector $\mathbf{h} = (h_1, \ldots, h_{J(k)+1}) \in \mathbb{Z}^{J(k)+1}$ with $0 < \|\mathbf{h}\|_\infty \leq H$ the phase is

$$\Psi_{\mathbf{h}}(x) = \sum_{i \leq J(k)+1} h_i \, n \, x^{-j_i}, \tag{14}$$

and $S(\mathbf{h}) = \sum_{p \in \text{band}} e(\Psi_{\mathbf{h}}(p))$ is the sum of (1). We write

$$i_0 = i_0(\mathbf{h}) := \max\{i : h_i \neq 0\}, \qquad j := j_{i_0} = k - i_0 + 1, \qquad F := |h_{i_0}| \, n \, Q^{-j}, \tag{15}$$

Q being the left endpoint of the block under consideration. By (11),

$$F = |h_{i_0}| \, t \, Q^{\,i_0 - 1}, \qquad \text{so} \qquad 1 \leq F \leq H \, Q^{\,J+1}, \qquad \text{and} \qquad F \geq Q \ \text{ whenever } i_0 \geq 2. \tag{16}$$

[A2] §7 speaks of "the amplitude $F_i = hn/Q^{k-i+1}$" as though each level contributed its own regime. The phase (14) is a *sum* of monomials, and what governs every derivative test is the single amplitude F of the *highest* non-vanishing level, (15); Lemma 21 below shows that the lower levels are dominated by a factor $Q^{-1}e^r$ and never interfere. Read this way the sentence is correct, and the range $F \in [h, hQ^i]$ displayed there is (16).

B.1 The r-th derivative test with a constant uniform in r

Lemma 10 as displayed carries N^{r-2} inside the first term. That statement is true — it is implied by the one below, since $N \geq 1$ — but it is too weak to support the "in particular" clause of Lemma 11: with $\lambda N^{r-2} = Fj^r/L^2$ and $F \leq L^{r-\eta}$ the first term is $\geq L$ as soon as $r \geq 3$, so no saving results. **[A3]** The correct first term is $N\lambda^{1/(2^r-2)}$, and Lemma 10 should be restated as follows.

Lemma 19 (r-th derivative test, sharp form)**.** *Let $r \geq 2$, let $\mathcal{I}$ be an interval containing $N \geq 1$ integers, and let $f \in C^r(\mathcal{I})$ satisfy $\lambda \leq |f^{(r)}| \leq A\lambda$ on $\mathcal{I}$ with $\lambda > 0$, $A \geq 1$. Then*

$$\Big| \sum_{l \in \mathcal{I}} e(f(l)) \Big| \leq C_0 \, A^{2^{2-r}} \Big(N\lambda^{1/(2^r-2)} + N^{1-2^{2-r}} \lambda^{-1/(2^r-2)} \Big),$$

where C_0 is an absolute constant — in particular independent of r.

Proof. Write $\mu_r = 1/(2^r-2)$, $\beta_r = 2^{2-r}$, and $B_r(N,\lambda) := N\lambda^{\mu_r} + N^{1-\beta_r}\lambda^{-\mu_r}$. Since $\lambda \leq |f^{(r)}|$ and $f^{(r)}$ is continuous, $f^{(r)}$ has constant sign on $\mathcal{I}$.

If $N \le 2$ the claim is trivial: the left side is ≤ 2, while $B_r \ge 2\sqrt{N \cdot N^{1-\beta_r}} = 2N^{1-\beta_r/2} \ge 2$ by AM–GM, so $C_0 \ge 1$ suffices. Assume $N \ge 2$.

Case $r = 2$. This is Lemma 8 with $c = A$; the proof given there yields $|S| \le c_2 A(N\lambda^{1/2} + \lambda^{-1/2})$ with c_2 absolute, and $A^{2^{2-2}} = A$, $\mu_2 = 1/2$, $\beta_2 = 1$.

Case $r \ge 3$, granting the statement for $r-1$. Two degenerate ranges first. If $\lambda \ge 1$ then $|S| \le N \le N\lambda^{\mu_r}$. If $\lambda^{-2\mu_r} \ge N$, then $\lambda \le N^{-1/(2\mu_r)} = N^{-(2^{r-1}-1)} \le N^{-(4-2^{3-r})} = N^{-\beta_r/\mu_r}$ (the middle inequality because $2^{r-1} - 1 \ge 4 - 2^{3-r}$ for $r \ge 3$, with equality at $r = 3$), whence $N^{1-\beta_r}\lambda^{-\mu_r} \ge N$ and again $|S| \le N$ suffices.

So assume $1 \le \lambda^{-2\mu_r} < N$ and set $\mathcal{H} := \lceil \lambda^{-2\mu_r} \rceil$, so $1 \le \mathcal{H} \le N$ and $\lambda^{-2\mu_r} \le \mathcal{H} \le 2\lambda^{-2\mu_r}$. Apply Lemma 9 with $H = \mathcal{H}$. The term $h = 0$ contributes $(2N/\mathcal{H}) \cdot N$. For $1 \le h < \mathcal{H}$ put $f_h(x) = f(x+h) - f(x)$; by the mean value theorem $f_h^{(r-1)}(x) = h\, f^{(r)}(\xi)$ for some $\xi \in (x, x+h)$, so $h\lambda \le |f_h^{(r-1)}| \le Ah\lambda$. The inductive hypothesis (with $\theta := \mu_{r-1}$, $\beta_{r-1} = 2\beta_r$) gives

$$\Big|\sum_l e(f_h(l))\Big| \le C_0 A^{2\beta_r}\Big(N(h\lambda)^{\theta} + N^{1-2\beta_r}(h\lambda)^{-\theta}\Big).$$

Since $r \ge 3$ we have $\theta = 1/(2^{r-1}-2) \le \frac{1}{2}$, so $\sum_{1\le h<\mathcal{H}} h^{\theta} \le 2\mathcal{H}^{1+\theta}$ and $\sum_{1\le h<\mathcal{H}} h^{-\theta} \le 2\mathcal{H}^{1-\theta}$. Therefore

$$|S|^2 \le \frac{2N^2}{\mathcal{H}} + 4C_0 A^{2\beta_r}\Big(N^2(\lambda\mathcal{H})^{\theta} + N^{2-2\beta_r}(\lambda\mathcal{H})^{-\theta}\Big).$$

Now the arithmetic identity that makes the choice of $\mathcal{H}$ exact:

$$\theta\,(1-2\mu_r) = \frac{1}{2^{r-1}-2}\cdot\frac{2^r-4}{2^r-2} = \frac{1}{2^{r-1}-2}\cdot\frac{2(2^{r-1}-2)}{2^r-2} = \frac{2}{2^r-2} = 2\mu_r.$$

With $\lambda\mathcal{H} \in [\lambda^{1-2\mu_r}, 2\lambda^{1-2\mu_r}]$ this gives $(\lambda\mathcal{H})^{\theta} \le 2\lambda^{2\mu_r}$ and $(\lambda\mathcal{H})^{-\theta} \le \lambda^{-2\mu_r}$, while $2N^2/\mathcal{H} \le 2N^2\lambda^{2\mu_r}$. Since $N^{2-2\beta_r} = (N^{1-\beta_r})^2$,

$$|S|^2 \le (2 + 12C_0A^{2\beta_r})\Big(N^2\lambda^{2\mu_r} + \big(N^{1-\beta_r}\big)^2\lambda^{-2\mu_r}\Big) \le \big(2+12C_0\big)A^{2\beta_r}\,B_r(N,\lambda)^2,$$

using $A \ge 1$ and $u^2 + v^2 \le (u+v)^2$. Taking square roots, $|S| \le (2+12C_0)^{1/2}A^{\beta_r}B_r(N,\lambda)$. Thus the constant obeys the recursion $C_r \le (2 + 12C_{r-1})^{1/2}$, whose map $u \mapsto (2+12u)^{1/2}$ is increasing with unique fixed point $u^* = 6 + \sqrt{38} < 13$; hence $C_r \le \max(c_2, 13) =: C_0$ for every r. □

Remark 20 (Trap check 1: the constant in r)**.** *The concern that the r-th derivative test carries a factor $2^{O(r)}$, with r growing in n, does not materialise: the induction step* squares *the sum before applying the inductive hypothesis and then takes a square root, so the constants satisfy $C_r = (2+12C_{r-1})^{1/2}$ and converge rather than compound. The constant is absolute.*

What does compound is the ratio *A, through the factor $A^{2^{2-r}}$. This factor is harmless only if $A = O(1)$, and on a* dyadic *block A is not $O(1)$: for the phase $x \mapsto nx^{-j}$ on $x \sim Q$ one has $|f^{(r)}(2Q)|/|f^{(r)}(Q)| = 2^{-(j+r)}$, so $A = 2^{j+r}$ with j as large as $k \le K = (\log n)^{1-\varepsilon_n}$. Then $A^{2^{2-r}} = \exp\big(4(j+r)2^{-r}\log 2\big)$, which for $r \le r_{\max} = 3J+5$ and $j \asymp K$ is $\exp\big(\Omega\big(K\,2^{-3J}\big)\big) = \exp\big(\Omega\big((\log n)^{1-\varepsilon_n}e^{-O((\log\log n)^{1/3})}\big)\big)$ — larger than any power of $\log n$, and larger than the saving. This is the real content of the trap, and it is why §B.3 replaces the dyadic block by the block of ratio $\rho = 1 + 1/\varrho$.* ***[A4]***

For completeness we also record the verification demanded of the $2^{O(r)}$ form, which holds a fortiori: with $r \le r_{\max} = 3J+5$, $2^{O(r)} = \exp\bigl(O((\log\log n)^{1/3})\bigr)$, whereas the saving factor established in §B.7 is $\exp\bigl(c\,(\log n)^{\varepsilon_n} 2^{-3J-6}\bigr) = \exp\bigl(\exp\bigl((\log\log n)^{2/3}(1+o(1))\bigr)\bigr)$; the logarithm of the saving already exceeds the logarithm of $2^{O(r)}$ by an exponential factor.

B.2 Band phases and dominance

For $r \ge 1$ write $(j)_r := j(j+1)\cdots(j+r-1)$, so that $\frac{d^r}{dx^r}x^{-j} = (-1)^r (j)_r x^{-j-r}$. Note $(j)_r$, not j^r, is the exact factor; the two agree only up to $e^{O(r^2/j)}$, and for $j=1$ one has $(1)_r = r!$. **[A5]** Lemma 11 writes $g^{(r)}(l) \asymp_{j,r} Fj^r/L^r$ and then carries j^r through the displayed bound; the correct factor is $(j)_r$. Since Lemma 11 declares its implied constant to depend on j and r the statement is not false, but j and r both grow with n here, so the dependence must be made explicit; §B.4 does this.

Lemma 21 (Dominance)**.** *Let $\mathbf{h}$, i_0, j, F be as in (15), let $2 \le r \le r_{\max}$, and let $\mathcal{Q} = (Q, \rho Q]$ with $Q \ge 8e\,H\,e^{r_{\max}}$. Then for all $x \in \mathcal{Q}$,*

$$\frac{3}{4e}\,\frac{F\,(j)_r}{Q^r} \;\le\; \bigl|\Psi_{\mathbf{h}}^{(r)}(x)\bigr| \;\le\; \frac{5}{4}\,\frac{F\,(j)_r}{Q^r}.$$

In particular $\Psi_{\mathbf{h}}^{(r)}$ has constant sign on $\mathcal{Q}$ and $\lambda \le |\Psi_{\mathbf{h}}^{(r)}| \le A\lambda$ there with $\lambda := \frac{3}{4e}F(j)_r Q^{-r}$ and $A \le \frac{5e}{3} < 5$.

Proof. By (14), $\Psi_{\mathbf{h}}^{(r)}(x) = (-1)^r \sum_{i\le i_0} h_i n (j_i)_r x^{-j_i-r}$. Write $x^{-j_i-r} = Q^{-j_i-r}\theta_i(x)$ with $\theta_i(x) = (x/Q)^{-j_i-r}$. Since $1 \le x/Q \le \rho$ and $j_i + r \le k + r_{\max} = \varrho$,

$$\rho^{-\varrho} = \Bigl(1+\tfrac{1}{\varrho}\Bigr)^{-\varrho} \ge e^{-1} \quad\Longrightarrow\quad \theta_i(x) \in [e^{-1}, 1].$$

Put $G_i := |h_i|\,n\,(j_i)_r Q^{-j_i}$, so $G_{i_0} = F(j)_r$. For $i < i_0$ we have $j_i > j$ and

$$\frac{(j_i)_r}{(j)_r} = \prod_{s=0}^{r-1}\frac{j_i+s}{j+s} \le \Bigl(\frac{j_i}{j}\Bigr)^r = \Bigl(1+\frac{i_0-i}{j}\Bigr)^r \le e^{\,r(i_0-i)/j} \le e^{\,r(i_0-i)},$$

using that $s \mapsto (a+s)/(b+s)$ decreases for $a \ge b > 0$. Also $|h_i|nQ^{-j_i} = |h_i|\,t\,Q^{i-1} \le H\,t\,Q^{i-1}$ and $|h_{i_0}|nQ^{-j} = |h_{i_0}|tQ^{i_0-1} \ge tQ^{i_0-1}$, so

$$\sum_{i<i_0}\frac{G_i}{G_{i_0}} \le H\sum_{s\ge1}\Bigl(\frac{e^r}{Q}\Bigr)^s \le \frac{2He^r}{Q} \le \frac{1}{4e},$$

the last step by the hypothesis $Q \ge 8eHe^{r_{\max}}$ (which also gives $e^r \le Q/2$, justifying the geometric sum). Consequently

$$\bigl|\Psi_{\mathbf{h}}^{(r)}(x)\bigr|Q^r \;\ge\; e^{-1}G_{i_0} - \sum_{i<i_0} G_i \;\ge\; \Bigl(e^{-1} - \tfrac{1}{4e}\Bigr)G_{i_0} = \tfrac{3}{4e}F(j)_r,$$

and $\le (1+\frac{1}{4e})G_{i_0} \le \frac{5}{4}F(j)_r$. □

The hypothesis $Q \ge 8eHe^{r_{\max}}$ is satisfied for every block of every band with $k \le K$: $\log\bigl(8eHe^{r_{\max}}\bigr) = r_{\max} + 2\log\log n + O(1) = O(\log\log n)$, whereas

$$\log Q \;\ge\; \log P_0 = \frac{\log n}{k+1} \;\ge\; \frac{\log n}{(\log n)^{1-\varepsilon_n}+1} \;\ge\; \tfrac12(\log n)^{\varepsilon_n} = \tfrac12\exp\bigl((\log\log n)^{2/3}\bigr). \tag{17}$$

B.3 Fine blocks

Definition 22. A *fine block* of the band is an interval $\mathcal{Q} = (Q, \rho Q]$ contained in $(P_0, P]$, with $\rho = 1 + 1/\varrho$ as in (13).

The band is covered by

$$\mathcal{N}_{\mathrm{bl}} = \left\lceil \frac{\log(P/P_0)}{\log \rho} \right\rceil \;\le\; 1 + 2\varrho\, \frac{\log n}{k(k+1)}$$

fine blocks (using $\log \rho \ge \frac{1}{2\varrho}$), and every fine block contains $\asymp Q/\varrho$ integers. All estimates below are proved for one fine block and are *proportional to the block length*, so summing over blocks reproduces the band with no loss: if $|S_{\mathcal{Q}}(\mathbf{h})| \le |\mathcal{Q}|\,\Xi$ for every fine block with a saving factor Ξ independent of the block, then

$$|S(\mathbf{h})| \le \sum_{\mathcal{Q}} |\mathcal{Q}|\,\Xi \le P\,\Xi. \tag{18}$$

Since $P \le \pi(P)\log P$, a saving $\Xi \ll (\log n)^{-B-1}$ gives $|S(\mathbf{h})| \ll \pi(P)\log P\,(\log n)^{-B-1} \ll \pi(P)(\log n)^{-B}$, which is (1). The saving is measured against $\log n$ and not $\log Q$ or $\log P$; §B.8 supplies it in that form, and §B.9 shows why the weaker $\log P$-form would not close the argument ([**B2**]).

B.4 Type I on a fine block: the adaptive choice of r

Lemma 23 (Type I, effective form)**.** *Let $\mathcal{Q} = (Q, \rho Q]$ be a fine block, $m \ge 1$, and let $\mathcal{L}$ be an interval of l with $ml \in \mathcal{Q}$, containing $N \ge 2$ integers; put $L := Q/m$. Let $\mathbf{h}$, j, F be as in* (15) *and let $g(l) := \Psi_{\mathbf{h}}(ml)$. Put*

$$x := \frac{\log F}{\log N}, \qquad r := \max\bigl(2, \lceil x \rceil + 1\bigr), \qquad \varsigma := \frac{r\log\varrho + 8r}{\log N}.$$

Assume $2 \le r \le r_{\max}$ and $\varsigma \le \min\bigl(\frac{1}{16}, x/4\bigr)$ — the latter holds as soon as $F \ge e^{64}\varrho^8$. Then

$$\Bigl|\sum_{l\in\mathcal{L}} e(g(l))\Bigr| \;\ll\; N^{1-\eta'}, \qquad \eta' := \begin{cases} x/4, & r = 2,\\ \dfrac{1}{2(2^r-2)} \;\ge\; 2^{-r_{\max}-1}, & r \ge 3,\end{cases}$$

with an absolute implied constant. For $r = 2$ the conclusion reads

$$\Bigl|\sum_{l\in\mathcal{L}} e(g(l))\Bigr| \;\ll\; N\,F^{-1/4}:$$

the saving is a power of the amplitude alone, independent of N. This is the point that determines where the nearly stationary regime really begins; see §B.6.

Proof. Since $ml \in \mathcal{Q}$ and m is fixed, $\mathcal{L} \subseteq (L, \rho L]$, and $g(l) = \sum_i (h_i n m^{-j_i}) l^{-j_i}$ is a phase of the shape (14) with n replaced by nm^{-j_i} at level i and Q replaced by L. The amplitude of the top level is unchanged: $|h_{i_0}|nm^{-j}L^{-j} = |h_{i_0}|n(mL)^{-j} = |h_{i_0}|nQ^{-j} = F$; likewise every ratio G_i/G_{i_0} in the proof of Lemma 21 is unchanged, because $m^{-j_i}L^{-j_i} = Q^{-j_i}$ for every i. Hence Lemma 21 applies verbatim on $\mathcal{L}$ and gives

$$\lambda \le |g^{(r)}| \le A\lambda \ \text{ on } \mathcal{L}, \qquad \lambda = \tfrac{3}{4e}F(j)_r L^{-r}, \qquad A < 5.$$

Set $\kappa := \lambda N^r$. Since $\mathcal{L} \subseteq (L, \rho L]$ we have $N \le L/\varrho + 1 \le 2L/\varrho$ and $N \ge 1$, so $(N/L)^r \in [\,L^{-r},\, (2/\varrho)^r]$, and since $1 \le (j)_r \le (k+r)^r \le \varrho^r$,

$$\kappa = \tfrac{3}{4e}F\,(j)_r\,(N/L)^r \quad \Longrightarrow \quad \bigl|\log\kappa - \log F\bigr| \le r\log\varrho + 8r, \quad \text{i.e.} \quad \bigl|\log_N \kappa - x\bigr| \le \varsigma \le \tfrac{1}{16}.$$

Apply Lemma 19. With $\mu_r = 1/(2^r-2)$ and $\eta' = \mu_r/2$ the two terms are $\le N^{1-\eta'}$ provided

$$N\lambda^{\mu_r} \le N^{1-\eta'} \iff \kappa \le N^{\,r-\eta'/\mu_r} = N^{\,r-1/2}, \tag{19}$$

$$N^{1-2^{2-r}}\lambda^{-\mu_r} \le N^{1-\eta'} \iff \kappa \ \ge \ N^{\,r-(2^{2-r}-\eta')/\mu_r} = N^{\,r-4+2^{3-r}+1/2}. \tag{20}$$

By the displayed slack it suffices that $x \le r - \frac12 - \varsigma$ and $x \ge r - \frac72 + 2^{3-r} + \varsigma$.

If $r = 2$ (so $x \le 1$), take $\eta' := x/4$; then (19) and (20) read $\kappa \le N^{2-x/2}$ and $\kappa \ge N^{x/2}$, both implied by $|\log_N \kappa - x| \le \varsigma \le x/4$ together with $x \le 1$. Here $N^{1-\eta'} = N\,N^{-x/4} = N\,F^{-1/4}$, which is the second displayed form.

If $r \ge 3$, then $r = \lceil x\rceil + 1$ gives $x \le r-1$ and $x \ge r-2$. The upper condition needs $x \le r - \frac12 - \varsigma$: true since $x \le r-1$. The lower condition needs $x \ge r - \frac72 + 2^{3-r} + \varsigma$; for $r = 3$ this is $x \ge -\frac12 + 1 + \varsigma = \frac12 + \varsigma$, and $x \ge r-2 = 1$; for $r \ge 4$, $2^{3-r} \le \frac12$ so it is $x \ge r - 3 + \varsigma$, and $x \ge r-2$. Both hold. Finally $\eta' = \mu_r/2 \ge 2^{-r_{\max}-1}$ for $r \le r_{\max}$. □

Remark 24 (The window structure)**.** *The mechanism is worth stating separately, since it is what the "in particular" clause of Lemma 11 was reaching for. For fixed r the derivative test saves exactly on the window*

$$\kappa \in \bigl[N^{\,r-4+2^{3-r}+\frac12},\ N^{\,r-\frac12}\bigr],$$

of width $3.5 - 2^{3-r}$ *in the exponent. Consecutive windows overlap (*r *and* $r+1$ *share an exponent range of length* ≥ 2.5*), and their union over* $2 \le r \le r_{\max}$ *is* $[N^{\eta}, N^{\,r_{\max}-1/2}]$ *for any fixed* $\eta > 0$ *and n large. **[A3, continued]** Lemma 11 asserts a saving on* $L^\eta \le F \le L^{r-\eta}$ for a fixed r*; that is false for* $r \ge 5$*, where the lower edge of the window is at* $F \asymp L^{r-3.5}$*, not* L^η*. The correct statement is:* r *must be chosen as a function of* F *and* L*, namely* $r = \max(2, \lceil \log F/\log L\rceil + 1)$*, and then the saving exponent is* $\eta' \asymp 2^{-r}$*. The paper's* $r = J+3$ *is the* maximum r *needed, not a fixed choice; see **[A6]**.*

B.5 Type II on a fine block: the differenced amplitude

Lemma 25 (Trap check 2: the differenced amplitude)**.** *Let* $j \ge 1$*, let* $\mathcal{L} \subseteq (L, \rho L]$ *with* $\rho = 1 + 1/\varrho$*,* $\varrho \ge j$*, and let* $l, l' \in \mathcal{L}$*,* $\Delta := l' - l \ne 0$*. Then*

$$\bigl|l^{-j} - l'^{-j}\bigr| \ \asymp\ \frac{j\,|\Delta|}{L}\,L^{-j},$$

with absolute implied constants. If instead l, l' *are allowed to range over a full dyadic block* $(L, 2L]$*, then* $|l^{-j} - l'^{-j}| \asymp \min(l,l')^{-j}\min\bigl(1, j|\Delta|/L\bigr)$*, and the two sides of the first display differ by a factor as large as* 2^j*.*

Proof. By the mean value theorem $l^{-j} - l'^{-j} = j\Delta\, \xi^{-j-1}$ for some ξ between l and l'. On a fine block $\xi \in (L, \rho L]$ and $\rho^{-(j+1)} \geq \rho^{-\varrho} \geq e^{-1}$, so $\xi^{-j-1} \asymp L^{-j-1}$, giving the first claim. On a dyadic block write $l^{-j} - l'^{-j} = l^{-j}\big(1-(1+\Delta/l)^{-j}\big)$ with $|\Delta/l| \leq 1$; the bracket is $\asymp j|\Delta|/l$ when $j|\Delta|/l \leq 1$ and $\asymp 1$ otherwise, while l^{-j} ranges over $[(2L)^{-j}, L^{-j}]$. $\square$

[A6] Lemma 13 states that the differenced phase $h\,n\,m^{-j}(l^{-j} - l'^{-j})$ is "a monomial in m of amplitude $\asymp F|l-l'|/L$". Two corrections are needed. *(a) The factor j is missing*: by Lemma 25 the amplitude is $\asymp F\,j|l-l'|/L$. *(b) The estimate is a fine-block statement*: over the dyadic l-range that Lemma 13 allows, it is false by a factor up to 2^j, and 2^j can be as large as 2^K. Both corrections are incorporated below. A third point: *(c)* the proof applies Lemma 11 *in the variable* m, so the saving is a power of M, not of L; the displayed conclusion $Q(\log Q)^3(M^{-1/2} + L^{-\eta'/2})$ should read $Q(\log Q)^3(L^{-1/2} + M^{-\eta'/2})$. (The first summand as printed is weaker than what the proof gives, hence harmless; the second as printed is *stronger*, hence an error. In the application both are far below the target.) Finally *(d)*: the hypothesis "$L^\eta \leq F|\Delta|/L$ for $1 \leq |\Delta| \leq L$" forces $F \geq L^{1+\eta}$, which fails throughout the shallow part of every band; Lemma 26 replaces it by a count of the exceptional pairs.

Lemma 26 (Type II, effective form)**.** *Let $\mathcal{Q} = (Q, \rho Q]$ be a fine block, let $M \geq 2$ with $L := Q/M \geq Q^{1/3}$, and let*

$$\mathcal{B} = \sum_{m\sim M}\ \sum_{l\,:\,ml\in\mathcal{Q}} a_m b_l\, e\big(\Psi_{\mathbf{h}}(ml)\big), \qquad |a_m| \leq \tau(m),\ |b_l| \leq \log l.$$

Let $\mathbf{h}, j, F$ be as in (15) *and assume $Fj \geq e^{80}\varrho^{10}$, and $r \leq r_{\max}$ for every r arising below. Then*

$$|\mathcal{B}| \ \ll\ Q(\log Q)^{5/2}\Big(L^{-1/2} + (Fj)^{-1/10} + M^{-2^{-r_{\max}-2}}\Big),$$

with an absolute implied constant.

Proof. The l's occurring satisfy $l \in (Q/(2M), \rho Q/M]$, a range of multiplicative width $2\rho \leq 3$. Split it into $\nu_{\max} \leq 1 + \varrho\log 3 \ll \varrho$ fine blocks $\mathcal{L}_\nu = (L_\nu, \rho L_\nu]$ with $L_\nu \asymp L$, and write $\mathcal{B} = \sum_\nu \mathcal{B}_\nu$ accordingly. Fix ν. Given $l \in \mathcal{L}_\nu$ and $ml \in \mathcal{Q}$, the variable m lies in $\mathcal{M}_\nu := (Q/(\rho L_\nu), \rho Q/L_\nu]$, of multiplicative width ρ^2; put $N_m := \#(\mathcal{M}_\nu \cap \mathbb{Z}) \asymp M/\varrho$. By Cauchy–Schwarz in m and Lemma 15,

$$|\mathcal{B}_\nu|^2 \leq \Big(\sum_{m\sim M}\tau(m)^2\Big)\sum_{m\in\mathcal{M}_\nu}\Big|\sum_{\substack{l\in\mathcal{L}_\nu\\ ml\in\mathcal{Q}}} b_l e(\Psi_{\mathbf{h}}(ml))\Big|^2 \ll M(\log 2M)^3\,\Sigma,$$

where, expanding the square,

$$\Sigma = \sum_{l,l'\in\mathcal{L}_\nu} b_l\overline{b_{l'}} \sum_{m\in\mathcal{M}_\nu(l,l')} e\big(\Psi_{\mathbf{h}}(ml) - \Psi_{\mathbf{h}}(ml')\big),$$

and $\mathcal{M}_\nu(l,l') = \{m \in \mathcal{M}_\nu : ml, ml' \in \mathcal{Q}\}$ is an interval. Since $|b_l| \leq \log(2L) \leq \log Q$, the diagonal $l = l'$ contributes $\ll (\log Q)^2|\mathcal{L}_\nu|N_m \ll (\log Q)^2 Q/\varrho^2$.

For $l \neq l'$ the inner phase is $\Psi_{\mathbf{h}}(ml) - \Psi_{\mathbf{h}}(ml') = \sum_i h_i n(l^{-j_i} - l'^{-j_i})m^{-j_i}$, a phase of the shape (14) in m whose level-i coefficient is $h_i n(l^{-j_i} - l'^{-j_i})$. By Lemma 25, the top-level amplitude at scale M is

$$G := \big|h_{i_0}\big|\, n\,\big|l^{-j} - l'^{-j}\big|\Big(\frac{Q}{L_\nu}\Big)^{-j} \asymp F\,\frac{j|l-l'|}{L},$$

and the dominance estimate of Lemma 21 holds with the same proof (the ratios G_i/G_{i_0} pick up the extra factor $\left|l^{-j_i} - l'^{-j_i}\right| / \left|l^{-j} - l'^{-j}\right| \cdot L^{j_i - j} \le j_i/j \le k$, absorbed by the margin $Q \ge 8eHKe^{r_{\max}}$, which (17) again grants).

Put $G_0 := (Fj)^{4/5}$ and call a pair (l, l') *good* if $G \ge G_0$. For a good pair, Lemma 23 applied in the variable m (with F replaced by G, L by M, N by N_m) gives $|\sum_m e(\cdot)| \ll N_m \Xi$, where

$$\Xi := G_0^{-1/4} + N_m^{-2^{-r_{\max}-1}} :$$

if the resulting exponent x is ≤ 1 the saving is $G^{-1/4} \le G_0^{-1/4}$, and otherwise it is $N_m^{-\eta'}$ with $\eta' \ge 2^{-r_{\max}-1}$. The hypothesis $G \ge e^{64}\varrho^8$ of Lemma 23 holds because $G \ge G_0 = (Fj)^{4/5} \ge (e^{80}\varrho^{10})^{4/5} = e^{64}\varrho^8$. For a bad pair we use the trivial bound N_m; by Lemma 25 a pair is bad only if $|l - l'| \ll G_0 L/(Fj)$, so the number of bad pairs is $\ll |\mathcal{L}_\nu|\bigl(1 + G_0 L/(Fj)\bigr)$. Hence

$$\Sigma \ll (\log Q)^2 \Big[\frac{Q}{\varrho^2} + \frac{L^2}{\varrho^2} N_m \Xi + \frac{L}{\varrho}\Big(1 + \frac{G_0 L}{Fj}\Big) N_m\Big],$$

and therefore, using $N_m \asymp M/\varrho$ and $ML = Q$,

$$|\mathcal{B}_\nu|^2 \ll M(\log Q)^5 \Big[\frac{Q}{\varrho^2} + \frac{Q^2}{M\varrho^3}\Xi + \frac{Q}{\varrho^2}\Big(1 + \frac{G_0 L}{Fj}\Big)\Big].$$

Taking square roots, summing over the $\ll \varrho$ values of ν, and using $(MQ)^{1/2} = QL^{-1/2}$,

$$|\mathcal{B}| \ll (\log Q)^{5/2}\, Q \Big[L^{-1/2} + \Xi^{1/2} + \Big(\frac{G_0}{Fj}\Big)^{1/2}\Big].$$

The exponent $4/5$ is chosen to balance the last two: both $G_0^{-1/8}$ and $(G_0/(Fj))^{1/2}$ equal $(Fj)^{-1/10}$. Finally $N_m^{-2^{-r_{\max}-2}} \le 2M^{-2^{-r_{\max}-2}}$ because $2^{-r_{\max}-2}\log\varrho = o(1)$. $\square$

Remark 27. *The bound is* $\ll Q(\log Q)^{-A}$ *as soon as*

$$Fj \ge (\log Q)^{10A+40} \qquad \text{and} \qquad M \ge (\log Q)^{(A+3)2^{r_{\max}+2}},$$

the second being automatic in §B.7 by (17). The bilinear estimate therefore reaches down to amplitudes that are a fixed power of $\log Q$, *not merely to* $F \ge L^{1+\eta}$ *as the hypothesis of Lemma 13 would require, nor to* $F \ge L^\eta$ *as §7 suggests. This is what makes §B.6 possible.*

Lemma 28 (Symmetric range)**.** *Lemma 26 holds verbatim with the roles of m and l exchanged, i.e. for $|a_m| \le \log m$, $|b_l| \le \tau(l)$, with M and L exchanged throughout. Consequently the bilinear range $U/2 \le M \le Q/U$ of Lemma 16 may be reduced to $U/2 \le M \le Q^{1/2}$.*

Proof. The proof of Lemma 26 uses about a only $\sum_{m\sim M} |a_m|^2 \ll M(\log 2M)^3$ and about b only $|b_l| \le \log Q$; both hold after the exchange, the first by Lemma 15 applied in l and the second because $|a_m| \le \log m \le \log Q$. The phase $\Psi_{\mathbf{h}}(ml)$ is symmetric in m and l. For the reduction: if $M > Q^{1/2}$ then $L = Q/M < Q^{1/2}$, and applying the exchanged form to $\mathcal{B}$ with outer variable l gives the bound with M and L interchanged. $\square$

B.6 Where the nearly stationary regime really begins

By (16), $F \geq Q$ whenever $i_0 \geq 2$. Small amplitudes therefore occur only for frequency vectors supported at level one, where $F = |h_1|\,t$ with $1 \leq t \leq P_0$. Fix the target saving $(\log n)^{-B}$ of (1) and set

$$T_0 := e^{80}\varrho^{10}(\log n)^{10B+40}. \tag{21}$$

Since $\varrho = k + r_{\max} \leq 3\log n$, we have $T_0 \leq (\log n)^{10B+50}$, a fixed power of $\log n$.

(a) $F \geq T_0$ is covered by the derivative tests. For Type I, $F \geq T_0 \geq e^{64}\varrho^8$ so Lemma 23 applies; if $x \leq 1$ its saving is $F^{-1/4} \leq (\log n)^{-(10B+40)/4} \leq (\log n)^{-B-2}$, and if $x > 1$ the saving is $N^{-\eta'}$ with $\eta' \geq 2^{-r_{\max}-1}$, which is $\leq (\log n)^{-B-2}$ by §B.8. For Type II, $Fj \geq F \geq e^{80}\varrho^{10}$ so Lemma 26 applies, and $(Fj)^{-1/10} \leq T_0^{-1/10} \leq (\log n)^{-B-4}$ by (21).

[A8′] Hence *the nearly stationary regime is* $F < T_0 = (\log n)^{O_B(1)}$, *not* $F \leq L^\eta$, and the difference is decisive rather than cosmetic. The mechanism is the last display of Lemma 23: at $r = 2$ the second-derivative test saves $F^{-1/4}$, a power of the amplitude that does not degrade as the interval lengthens, so a saving of a fixed power of $\log Q$ — all this paper asks for — is already available at F a fixed power of $\log Q$. Were the threshold really L^η, the direct count of part (b) would require primes counted in intervals of relative length $P^{-\eta/3}$, which is not available unconditionally; at T_0 it is routine.

(b) $F < T_0$: the direct count. Here $i_0 = 1$ and, since $|h_1| \geq 1$, also $t \leq F < T_0$. The map $p \mapsto t = n/p^k$ is a decreasing bijection of the band onto $[1, P_0]$, and the level-one phase $h_1 n/p^k = h_1 t$ is monotone in p with total variation $|h_1|T_0 \ll F + 1$ over $\{t \leq T_0\}$. Consequently, for any target arc I_1, the condition $\{n/p^k\} \in I_1$ meets $\{t \leq T_0\}$ in $O(F+1)$ intervals of p — this is the interval count referred to in §7. Writing $\mathcal{R}_\nu := \{p : t(p) \in (\nu, \nu+1]\}$ for $\nu \leq \lceil T_0 \rceil$, each $\mathcal{R}_\nu$ is an interval of p of multiplicative width $(1+1/\nu)^{1/k}$, hence of length

$$|\mathcal{R}_\nu| \asymp \frac{P}{k\nu} \;\geq\; \frac{P}{kT_0} \;\geq\; P\,(\log n)^{-10A-51}. \tag{22}$$

On $\mathcal{R}_\nu$ the level-one coordinate is $\{t\} = t - \nu$, monotone in p, so $\{n/p^k\} \in I_1$ cuts $\mathcal{R}_\nu$ into one subinterval $\mathcal{R}_\nu(I_1)$, and by Lemma 2 (log-uniformity of the Mertens measure in the coordinate t)

$$\frac{\sum_{p \in \mathcal{R}_\nu(I_1)} 1/p}{\sum_{p \in \mathcal{R}_\nu} 1/p} = \sigma_\nu(I_1)\big(1 + o(1)\big), \qquad \sigma_\nu(I_1) := \frac{\log \frac{\nu+\beta}{\nu+\alpha}}{\log \frac{\nu+1}{\nu}}, \quad I_1 = [\alpha, \beta). \tag{23}$$

Within $\mathcal{R}_\nu(I_1)$ the remaining coordinates $i = 2, \ldots, J(k)$ have $F \geq Q$ by (16) and are therefore covered by part (a); by (22) that interval still has $\log Q \asymp \log P$, so (17) is unaffected.

(c) What (b) costs, and the one input the paper does not list. Statement (23), and the band-Mertens statement $\sum_{p \in \text{band}} 1/p = \log(1 + 1/k)(1 + o(1))$ of component (B), both require the prime number theorem with the classical error term $\pi(y) = \mathrm{li}(y) + O\big(y\, e^{-c\sqrt{\log y}}\big)$. They do *not* follow from Chebyshev–Mertens alone: Mertens' theorem with its elementary error gives

$$\sum_{p \in \text{band}} \frac{1}{p} = \log\Big(1 + \frac{1}{k}\Big) + O\Big(\frac{k}{\log n}\Big),$$

whose error exceeds the main term $\asymp 1/k$ as soon as $k > \sqrt{\log n}$, whereas the schedule runs to $K = (\log n)^{1-\varepsilon_n} \gg \sqrt{\log n}$. With the classical error term the relative error in

(22)–(23) is $\ll kT_0\, e^{-c\sqrt{\log P}}$, and by (17) $\sqrt{\log P} \geq (\log n)^{\varepsilon_n/2} = \exp\big(\frac{1}{2}(\log\log n)^{2/3}\big)$ while $\log(kT_0) = O(\log\log n)$; the margin is overwhelming.

[A10] The input list of §3 — "Kummer's carry criterion ..., the Chebyshev–Mertens bounds of §2 (machine-verified in finite form), and the elementary analysis of §§5–6" — is therefore incomplete: it must also name the prime number theorem with a classical error term. This is an internal inconsistency rather than a new hypothesis: Remark 5 already reasons with "PNT-with-classical-error", and that theorem is unconditional. It is not, however, elementary in the sense the input list suggests, and it is not among the machine-verified components.

(d) A route that avoids (b) and (c) entirely. The theorem needs only an *upper* bound on the carry-free mass, so any one of the $J(k)$ level conditions may be discarded. Discarding level one and using the levels $i = 2, \ldots, J(k)+1$ instead, every frequency vector has $i_0 \geq 2$, hence $F \geq Q$ by (16): part (b) never arises, (23) is not needed, and the carry-free proportion of the band is $\leq 2^{-J(k)}(1+o(1))$ as before. The only cost is the requirement $J(k)+1 \leq k$, absorbed into (12). The band-Mertens input of (c) is still needed — it is what gives the band its mass $\asymp 1/k$ — but no statement about primes in short intervals is.

[A7] Independently of which route is taken, *Proposition 4 is false as stated at level one*: its conclusion asserts the density $\prod_i |I_i|$, whereas (23) — which is Lemma 2, confirmed numerically in the remark following it, and asserted in §2 to be "an exact statement, not a model" — gives $\sigma_\nu(I_1) \neq |I_1|$ at bounded ν. The two statements of the paper contradict each other. The correct form is

$$\#\{p \in \mathcal{R}_\nu : \{n/p^{k-i+1}\} \in I_i\ (1 \leq i \leq J(k))\} = \Big(\sigma_\nu(I_1) \prod_{2 \leq i \leq J(k)} |I_i| + o(1)\Big)\#\mathcal{R}_\nu,$$

uniformly for $\nu \leq T_0$, together with Proposition 4 unmodified for $t > T_0$. For §8 only the upper bound matters, and $\sigma_\nu([0,\frac{1}{2})) \leq \sigma_1 = \log(3/2)/\log 2 = 0.5850\ldots < 1$ (decreasing to $\frac{1}{2}$), so the carry-free proportion is $\leq 1.171 \cdot 2^{-J(k)}$ — a change of absolute constant only.

B.7 The case table

Fix a band $k \leq K$, a fine block $\mathcal{Q} = (Q, \rho Q]$, a level index $i \leq J(k)$, and a frequency $\mathbf{h}$ with $0 < \|\mathbf{h}\|_\infty \leq H$. Set $U := Q^{1/3}$ in Lemma 16, so that the Type I range is $M \leq U^2 = Q^{2/3}$, i.e. $L = Q/M \geq Q^{1/3}$, and the bilinear range is $Q^{1/3}/2 \leq M \leq Q^{2/3}$, reduced by Lemma 28 to $Q^{1/3}/2 \leq M \leq Q^{1/2}$. Then $x = \log F/\log N$ with $\log N \geq \frac{1}{3}\log Q(1-o(1))$ in both families, so by (16)

$$0 < x \leq \frac{\log H + (J+1)\log Q}{\frac{1}{3}\log Q(1-o(1))} \leq 3J{+}4, \qquad \text{hence} \qquad r = \max(2, \lceil x\rceil{+}1) \leq 3J{+}5 = r_{\max}. \tag{24}$$

Both hypotheses on the slack parameter $\varsigma = r(\log\varrho + 8)/\log N$ hold. For $r \geq 3$, $\varsigma \leq \dfrac{r_{\max}(\log\varrho+8)}{\frac{1}{3}\log Q(1-o(1))} \ll \dfrac{(\log\log n)^{4/3}}{(\log n)^{\varepsilon_n}} = o(1) \leq \frac{1}{16}$ by (17). For $r = 2$ the requirement $\varsigma \leq x/4$ is exactly $\log F \geq 8\log\varrho + 64$, i.e. $F \geq e^{64}\varrho^8$, which holds throughout the range $F \geq T_0$ of §B.6(a) by (21).

Under the top-level exclusion of Proposition 4 — the route of §B.6(d), and the one the paper takes — only row 1 of the table below arises, since every frequency vector then has

$i_0 \geq 2$ in the indexing of §B.0 (Remark 18). Rows 2 and 3 are retained for two reasons: the enumeration is exhaustive only with them, and they are what the corroborative top-level analysis of §2 requires.

#	regime	amplitude F	r used	covering statement
1	$i_0 \geq 2$, any t	$Q \leq F \leq HQ^{J+1}$	$\max(2, \lceil x \rceil + 1) \leq 3J + 5$	Lem. 23, 26
2	$i_0 = 1$, $F \geq T_0$	$T_0 \leq F \leq HQ$	$2 \leq r \leq 5$	Lem. 23, 26
3	$i_0 = 1$, $F < T_0$	$1 \leq F < (\log n)^{10B+50}$	—	§B.6(b), or void by (d)
sub-cases of 1–2, by the window of Remark 24:				
1a	$x \in (0, 1]$	$F \leq N$	$r = 2$	$\eta' = x/4$, saving $F^{-1/4}$
1b	$x \in (1, 2]$	$N < F \leq N^2$	$r = 3$	$\eta' = \frac{1}{12}$
1c	$x \in (2, 3]$	$N^2 < F \leq N^3$	$r = 4$	$\eta' = \frac{1}{28}$
1d	$x \in (s-1, s]$, $4 \leq s \leq 3J+4$	$N^{s-1} < F \leq N^s$	$r = s + 1$	$\eta' = \frac{1}{2(2^{s+1}-2)}$
Type II only: replace F by $G \asymp Fj\lvert l - l'\rvert/L$ and N by $N_m \asymp M/\varrho$; pairs with $G < G_0 = (Fj)^{4/5}$ are counted, not estimated (Lem. 26).				

Verification that cases 1–3 exhaust and overlap. Case 1 is $i_0 \geq 2$; cases 2 and 3 are $i_0 = 1$ with $F \geq T_0$ and $F < T_0$ respectively. These are disjoint and exhaustive. Case 2 has $F = |h_1|t \leq HP_0 \leq HQ$ by (11), so $x \leq 3(1 + \log H/\log Q) = 3 + o(1)$ and $r = \max(2, \lceil x \rceil + 1) \leq 5$ suffices. Case 1 has $F \geq Q$, so $x \geq 1$, and $F \leq HQ^{J+1}$, so $x \leq 3J + 4$. Sub-cases 1a–1d are the windows of Remark 24; by (24) they cover every $x > 0$ arising in cases 1–2 with $r \leq 3J + 5$, and the hypothesis $\varsigma \leq x/4$ needed in 1a is supplied by $F \geq T_0$. *No regime is uncovered.*

The Kusmin–Landau range. **[A8]** §7 describes the alternative to the derivative-test range as "the Kusmin–Landau range ($F \leq L^\eta$: the phase is nearly stationary...)". Lemma 7 requires $\|f'\| \geq \delta > 0$, which is exactly what a nearly stationary phase does not provide: for $F \leq L^\eta$ one has $|g'| \asymp Fj/L \ll 1$, so $\|g'\| = |g'|$ is small and Lemma 7 gives a bound $\gg L$, i.e. nothing. Kusmin–Landau enters this paper only as the engine inside the proof of Lemma 8, and it is Lemma 2 — the parenthetical's own content — that covers the nearly stationary regime. The sentence should name the band geometry, not Kusmin–Landau; and, by **[A8′]**, it should name the threshold T_0 of (21) rather than L^η.

B.8 The two displayed inequalities of §7

First display. With $r = r_{\max} = 3J + 5$ and $J = \lceil (\log\log n)^{1/3} \rceil$, Lemma 23 gives

$$\eta' \;\geq\; 2^{-r_{\max}-1} = \exp\bigl(-(3J+6)\log 2\bigr) \;\geq\; \exp\bigl(-c(\log\log n)^{1/3}\bigr), \qquad c := 5\log 2,$$

the last step being the inequality $(3J + 6)\log 2 \leq c(\log\log n)^{1/3}$, i.e. $3J + 6 \leq 5(\log\log n)^{1/3}$, which holds for n large because $J \leq (\log\log n)^{1/3} + 1$ gives $3J + 6 \leq 3(\log\log n)^{1/3} + 9$. **Verified**, with 2^{-J-3} replaced by 2^{-3J-6}; only the constant c changes. **[A6]** §7 takes "$r = J + 3$". The correct bound is $r \leq 3J + 5$: the Type I family runs down to $L = Q^{1/3}$ (Vaughan with $U = Q^{1/3}$ delivers $M \leq U^2 = Q^{2/3}$), so $x = \log F/\log N$ can be three times $\log F/\log Q \leq J + 1 + o(1)$, not once. The schedule is unaffected because 2^{-3J} and 2^{-J} are both $\exp(-O((\log\log n)^{1/3}))$.

Second display. For $k \leq K$, (17) gives $\log Q \geq \frac{1}{2}(\log n)^{\varepsilon_n}$, and $L \geq Q^{1/3}$ gives $\log L \geq \frac{1}{6}(\log n)^{\varepsilon_n}$. Hence

$$L^{\eta'} \;\geq\; \exp\Bigl(\tfrac{1}{6}(\log n)^{\varepsilon_n} e^{-c(\log\log n)^{1/3}}\Bigr),$$

which is the displayed inequality with $c' = \frac{1}{6}$. For the second half, $(\log n)^{\varepsilon_n} = \exp(\varepsilon_n \log\log n) = \exp\big((\log\log n)^{2/3}\big)$, so

$$\log\big(L^{\eta'}\big) \;\geq\; \tfrac{1}{6}\exp\Big((\log\log n)^{2/3} - c(\log\log n)^{1/3}\Big) \;=\; \exp\big((\log\log n)^{2/3}(1+o(1))\big),$$

whereas $\log\big((\log n)^A\big) = A\log\log n$. Since $\exp\big((\log\log n)^{2/3}(1+o(1))\big)/\log\log n \to \infty$, we get $L^{\eta'} \gg (\log n)^A$ for every fixed A, uniformly in $k \leq K$. **Verified.**

The trailing claim. "$\log L \gg \log Q \gg (\log n)^{\varepsilon_n} = \exp((\log\log n)^{2/3})$, which dominates $2^J = \exp(O((\log\log n)^{1/3}))$." The second $\gg$ and the identity are correct ((17) and the computation just made). The first $\gg$ is a typo: $\log L \leq \log Q$ always, and what holds is $\log L \geq \frac{1}{3}\log Q$, i.e. $\log L \asymp \log Q$. **[A9]** With that reading the sentence is correct, and it is the inequality that makes the whole schedule work: $\log Q$ is doubly exponential in $\log\log\log n$ while 2^J is only singly exponential in $(\log\log n)^{1/3}$.

B.9 Assembly of the appendix, and the flag list

Collecting: fix $k \leq K$ and $\mathbf{h}$ with $0 < \|\mathbf{h}\|_\infty \leq H$. On every fine block $\mathcal{Q}$ on which $F \geq T_0$ (cases 1–2 of §B.7), Lemma 16 with $U = Q^{1/3}$ reduces $\sum_{p\in\mathcal{Q}} e(\Psi_{\mathbf{h}}(p))$ to

$$\frac{Q}{U} + (\log Q)^2 \max_{M\leq Q^{2/3}} \mathcal{T}_M + (\log Q)^2 \max_{Q^{1/3}/2\leq M\leq Q^{1/2}} \mathcal{B}_M,$$

the last range by Lemma 28. The first term is $\leq |\mathcal{Q}|\cdot \varrho Q^{-1/3} \ll |\mathcal{Q}|(\log n)^{-B-1}$ by (17). For the second, Lemma 23 bounds each inner sum by $N^{1-\eta'}$ with $N \asymp L/\varrho$, so $\mathcal{T}_M \ll M\,(L/\varrho)^{1-\eta'} \ll |\mathcal{Q}|\,N^{-\eta'}$. For the third, Lemma 26 gives $\mathcal{B}_M \ll Q(\log Q)^{5/2}\big(L^{-1/2} + (Fj)^{-1/10} + M^{-2^{-r_{\max}-2}}\big)$; here $L^{-1/2} \leq Q^{-1/6}$, $(Fj)^{-1/10} \leq T_0^{-1/10} \leq (\log n)^{-B-4}$ by (21), and $M^{-2^{-r_{\max}-2}} \leq (\log n)^{-B-4}$ by §B.8 together with $M \geq Q^{1/3}/2$. By (17) every one of these is $\ll (\log n)^{-B-1}$, hence $|\sum_{p\in\mathcal{Q}} e(\Psi_{\mathbf{h}}(p))| \ll |\mathcal{Q}|(\log n)^{-B-1}$, and (18) yields (1). Where $F < T_0$, §B.6(b) applies instead — or, on the route of §B.6(d), that case does not occur.

Feed this into the Erdős–Turán inequality of Appendix C in dimension $J(k)$ with $H = (\log n)^2$ (Theorem 33 and Remark 34). Its first term is

$$\ll\; \pi(P)\,J\epsilon_H \;\ll\; \pi(P)(\log\log n)^{4/3}(\log n)^{-2} \;\ll\; \pi(P)(\log n)^{-1};$$

its outer factor J on the exponential-sum term is $\exp\big(O(\log\log\log n)\big)$; and the sum over $\mathbf{h}$ carries the weight

$$\sum_{0<\|\mathbf{h}\|_\infty\leq H} \prod_i (1+|h_i|)^{-1} \;\ll\; (3\log H)^J = \exp\big(O((\log\log n)^{1/3}\log\log\log n)\big),$$

which is dominated by the saving $\exp\big(-\exp((\log\log n)^{2/3}(1+o(1)))\big)$ of §B.8 with overwhelming margin.

It remains to check that $B = 2$ in (1) already suffices. The band contains $N \gg P/\log n$ primes while $\pi(P) \asymp Pk/\log n$, so $\pi(P)/N \ll k \leq K = (\log n)^{1-\varepsilon_n}$, and the exponential-sum term of Theorem 33, measured relative to N, is at most

$$J\,(3\log H)^J\,\frac{\pi(P)}{N}\,(\log n)^{-B} \;\ll\; \exp\big(u - u^{2/3} - Bu + O(u^{1/3}\log u)\big), \qquad u := \log\log n,$$

using $K = e^{u-u^{2/3}}$ and $J(3\log H)^J = e^{O(u^{1/3}\log u)}$. This is $\le e^{-u} = (\log n)^{-1}$ precisely when $(2-B)u \le u^{2/3} - O(u^{1/3}\log u)$, which at $B = 2$ reads $0 \le u^{2/3} - O(u^{1/3}\log u)$ and holds for n large because $u^{1/3} \gg \log u$. Together with the first term this gives Proposition 4 — as amended by [**A7**] — with $o(1) \le (\log n)^{-1}$, uniformly in $k \le K$.

Flags. [**A1**] J must be $J(k) = \min(J, k-1)$ (§B.0). [**A2**] the amplitude is that of the top non-vanishing level, not one per level (§B.0). [**A3**] Lemma 10's first term, and the "in particular" range of Lemma 11 (§B.1, Remark 24). [**A4**] dyadic blocks must be refined to ratio $1 + 1/\varrho$ (Remark 20). [**A5**] $(j)_r$, not j^r (§B.2). [**A6**] the differenced amplitude, the Type II saving variable, its hypothesis, and $r = J+3 \to r \le 3J+5$ (§B.5, §B.8). [**A7**] Proposition 4 versus Lemma 2 at level one (§B.6). [**A8**] the "Kusmin–Landau range" is not one, and [**A8′**] its threshold is $T_0 = (\log n)^{O(1)}$, not L^η (§B.6(a), §B.7). [**A9**] $\log L \gg \log Q$ should be $\log L \asymp \log Q$ (§B.8). [**A10**] the input list of §3 must also name the prime number theorem with a classical error term (§B.6(c)).

Two further items, raised by the re-indexing and now resolved in the text.

[**B1**] **The amplitude range under the top-level exclusion.** *Resolved:* the operative level set is $i = 2, \dots, J(k)+1$ (Remark 18), so (16) carries $F \le H\,Q^{\,J+1}$ and (13) carries $r_{\max} = 3J+5$, with $\varrho = k+3J+5$, $\eta' \ge 2^{-3J-6}$ and $c = 5\log 2$ in §B.8 — the choice recorded in the last sentence of (12) and propagated through (24), row 1 of the table, and §§3, 7.

[**B2**] **The target exponent in** (1)**.** *Resolved:* the saving is measured against $\log n$, not $\log P$ — (1) now reads $S(\mathbf{h}) \ll \pi(P)(\log n)^{-B}$ for every fixed B, with $(\log n)$ likewise in (21), in the closing sentence of §B.3 and in the bookkeeping above, and $B = 2$ shown sufficient there. A saving in $\log P$ alone would not close the argument: with $\log P = (\log n)^{\varepsilon_n} = e^{u^{2/3}}$, $u = \log\log n$, and $\pi(P)/N \ll k \le K = e^{u-u^{2/3}}$, the relative error from $(\log P)^{-A}$ is $\exp\bigl(u - (A+1)u^{2/3} + O(u^{1/3}\log u)\bigr)$, which reaches $(\log n)^{-1}$ only for $A \ge 2u^{1/3} - 1$ — no *fixed* A suffices. What §B.8 proves, $N^{\eta'} \ge \exp\bigl(\exp(u^{2/3}(1+o(1)))\bigr)$, is the $\log n$-measured form, so no estimate changes under the substitution.

No regime arising for $k \le K$, $i \le J(k)$, $\|\mathbf{h}\|_\infty \le H$ is left uncovered by §B.7; the flags above are corrections to statements, not gaps in coverage.

C A weak J-dimensional Erdős–Turán inequality

The inequality proved here is the weak one: log-power losses are tolerated, and the sharp Beurling–Selberg extremal functions are not used. What is proved is enough for §7, where the losses are absorbed with room to spare (§B.9).

Throughout, $\|y\|$ denotes distance to the nearest integer, $e(y) = e^{2\pi i y}$, functions on $\mathbb{R}/\mathbb{Z}$ are identified with 1-periodic functions on $\mathbb{R}$, and for $f \in L^1(\mathbb{R}/\mathbb{Z})$ we write $\hat{f}(h) = \int_0^1 f(y)e(-hy)\,dy$. An *arc* is a set $I = [\alpha, \beta) \subseteq \mathbb{R}/\mathbb{Z}$ of length $|I| = \beta - \alpha \in [0,1]$.

C.1 The one-dimensional approximants

Lemma 29. *For every arc I and every $h \in \mathbb{Z}$,* $\bigl|\widehat{\mathbf{1}_I}(h)\bigr| \le \dfrac{1}{1+|h|}$.

Proof. For $h = 0$ the left side is $|I| \le 1$. For $h \ne 0$, $\widehat{\mathbf{1}_I}(h) = \int_\alpha^\beta e(-hy)dy = \dfrac{e(-h\alpha) - e(-h\beta)}{2\pi i h}$, so the left side is $\leq \dfrac{1}{\pi|h|}$; and $\pi|h| \geq 1+|h|$ because $(\pi-1)|h| \geq \pi-1 > 1$. $\square$

Lemma 30 (Fejér-type kernel)**.** *Let $\theta > 0$, $s \in \mathbb{Z}_{\geq 1}$, and let $\phi_\theta := \theta^{-1}\mathbf{1}_{[-\theta/2,\theta/2]}$ on $\mathbb{R}$. Put $\Phi := \phi_\theta^{*2s}$ (the $2s$-fold convolution). Then $\Phi \geq 0$, $\int_{\mathbb{R}} \Phi = 1$, $\operatorname{supp}\Phi \subseteq [-s\theta, s\theta]$, and*

$$\hat{\Phi}(\xi) = \Big(\frac{\sin \pi\theta\xi}{\pi\theta\xi}\Big)^{2s}, \qquad \text{so} \qquad 0 \leq \hat{\Phi}(\xi) \leq \min\big(1, (\pi\theta|\xi|)^{-2s}\big).$$

Proof. Non-negativity, total mass and support are immediate from the corresponding properties of ϕ_θ and the fact that convolution adds supports. The Fourier transform of ϕ_θ is $\int_{-\theta/2}^{\theta/2} \theta^{-1} e(-\xi u)du = \frac{\sin \pi\theta\xi}{\pi\theta\xi}$, and convolution multiplies transforms. The final bound uses $|\sin| \leq 1$ and $|\sin z/z| \leq 1$. $\square$

The case $s = 1$ is the classical Fejér kernel (its transform is the triangle $(1-|\theta\xi|)_+$ up to rescaling); $s > 1$ is used only to make the transform decay fast enough that truncation at level H costs $O(1/H)$.

Proposition 31 (One-dimensional one-sided approximants)**.** *Let $H \geq 3$ be an integer and set*

$$s := \lceil \log H \rceil, \qquad \theta := \frac{2}{\pi H}, \qquad w := s\theta = \frac{2s}{\pi H}, \qquad \epsilon_H := 2w + \frac{1}{H} \leq \frac{4\log H + 6}{H}.$$

Assume H is large enough that $w \leq \frac{1}{8}$. Then for every arc I there are real trigonometric polynomials $\chi_I^{\pm}$ of degree $\leq H$ with

- *(i) $\chi_I^- \leq \mathbf{1}_I \leq \chi_I^+$ pointwise on $\mathbb{R}/\mathbb{Z}$ (hence also $\chi_I^+ \geq 0$);*
- *(ii) $\big|\hat{\chi}_I^{\pm}(0) - |I|\big| \leq \epsilon_H$;*
- *(iii) $\big|\hat{\chi}_I^{\pm}(h)\big| \leq \dfrac{1}{1+|h|}$ for $0 < |h| \leq H$.*

Proof. Let Φ be as in Lemma 30 with these θ, s, and periodize it; since $2w \leq \frac{1}{4}$ its periodization is supported in $[-w, w]$ modulo 1.

Majorant. If $|I| + 2w \geq 1$ put $\chi_I^+ := 1$; then (i) holds, (iii) is vacuous, and $\hat{\chi}^+(0) = 1 \leq |I| + 2w$, giving (ii). Otherwise let $I^+ := [\alpha - w, \beta + w)$, an arc of length $|I| + 2w < 1$, and $T^+ := \mathbf{1}_{I^+} * \Phi$. Then $0 \leq T^+ \leq 1$; and for $y \in I$ we have $y - u \in I^+$ whenever $|u| \leq w$, so $T^+(y) = \int \Phi = 1$. Thus $T^+ \geq \mathbf{1}_I$. Also $\hat{T}^+(h) = \widehat{\mathbf{1}_{I^+}}(h)\hat{\Phi}(h)$, so $\hat{T}^+(0) = |I| + 2w$ and, by Lemma 29 and Lemma 30, for $h \neq 0$

$$|\hat{T}^+(h)| \leq \frac{1}{\pi|h|}\,(\pi\theta|h|)^{-2s}.$$

In particular $|\hat{T}^+(h)| \ll |h|^{-2s-1}$, so the Fourier series of the continuous function T^+ converges absolutely to it. Put

$$\tau := \sum_{|h|>H} |\hat{T}^+(h)| \leq \frac{2}{\pi(\pi\theta)^{2s}} \sum_{h>H} h^{-2s-1} \leq \frac{2}{\pi(\pi\theta)^{2s}} \cdot \frac{1}{2s\,H^{2s}} = \frac{1}{\pi s}\,(\pi\theta H)^{-2s} = \frac{4^{-s}}{\pi s} \leq \frac{1}{H},$$

the last step because $\pi\theta H = 2$ and $4^{-s} \le 4^{-\log H} = H^{-\log 4} \le H^{-1}$. Now set $\chi_I^+ := \sum_{|h|\le H} \hat{T}^+(h)e(hy) + \tau$. It is a real trigonometric polynomial of degree $\le H$; it satisfies $\chi_I^+ \ge T^+ \ge \mathbf{1}_I$ because the omitted part of the series is at most τ in absolute value; $\hat{\chi}^+(0) = |I| + 2w + \tau$, which gives (ii); and $\hat{\chi}^+(h) = \hat{T}^+(h)$ for $0 < |h| \le H$, so (iii) follows from $|\hat{T}^+(h)| \le |\widehat{\mathbf{1}_{I^+}}(h)|$ and Lemma 29.

Minorant. If $|I| \le 2w$ put $\chi_I^- := 0$; then (i) and (iii) are clear and $|\hat{\chi}^-(0) - |I|| = |I| \le 2w \le \epsilon_H$. Otherwise let $I^- := [\alpha + w, \beta - w)$ and $T^- := \mathbf{1}_{I^-} * \Phi$. Then $0 \le T^- \le 1$ and $\operatorname{supp} T^- \subseteq I$, so $T^- \le \mathbf{1}_I$; and $\hat{T}^-(0) = |I| - 2w$. With τ^- defined as τ was, and by the same estimate $\tau^- \le 1/H$, set $\chi_I^- := \sum_{|h|\le H} \hat{T}^-(h)e(hy) - \tau^-$. Then $\chi_I^- \le T^- \le \mathbf{1}_I$, $\hat{\chi}^-(0) = |I| - 2w - \tau^-$, and (iii) holds as before. □

C.2 Products

Lemma 32 (Telescoping)**.** *Let a_ν, b_ν, c_ν be reals with $0 \le a_\nu \le c_\nu$ and $b_\nu \le a_\nu$ for $1 \le \nu \le J$. Then*

$$\prod_\nu c_\nu \;\ge\; \prod_\nu a_\nu \;\ge\; \prod_\nu c_\nu - \sum_\nu (c_\nu - b_\nu) \prod_{\mu \ne \nu} c_\mu.$$

Proof. The first inequality is monotonicity of the product of non-negative numbers. For the second, telescope:

$$\prod_\nu c_\nu - \prod_\nu a_\nu = \sum_{\nu=1}^{J} \Big(\prod_{\mu<\nu} a_\mu\Big)(c_\nu - a_\nu)\Big(\prod_{\mu>\nu} c_\mu\Big).$$

Every factor is non-negative, and $\prod_{\mu<\nu} a_\mu \le \prod_{\mu<\nu} c_\mu$, $c_\nu - a_\nu \le c_\nu - b_\nu$. □

C.3 The inequality

Theorem 33 (Weak J-dimensional Erdős–Turán)**.** *Let $J \ge 1$ and let $H \ge 3$ be an integer with $w \le \frac{1}{8}$ and $J\epsilon_H \le 1$, notation as in Proposition 31. Let $x_1, \dots, x_N \in [0,1)^J$ and let $B = \prod_{\nu \le J} I_\nu$ be a box, each I_ν an arc. Then*

$$\Big|\#\{i \le N : x_i \in B\} - N \operatorname{vol}(B)\Big| \;\le\; C\Big(NJ\,\epsilon_H + J \sum_{0<\|\mathbf{h}\|_\infty \le H} \; \prod_{\nu\le J} \frac{1}{1+|h_\nu|} \;\Big|\sum_{i\le N} e(\langle \mathbf{h}, x_i\rangle)\Big|\Big)$$

with C absolute. Since $\epsilon_H \ll (\log H)/H$, this is the asserted inequality up to the two log-power losses $\log H$ and J.

Proof. Write $\chi_\nu^\pm := \chi_{I_\nu}^\pm$, $A := \#\{i : x_i \in B\} = \sum_i \prod_\nu \mathbf{1}_{I_\nu}(x_{i,\nu})$, and for $\mathbf{h} \in \mathbb{Z}^J$, $\Sigma(\mathbf{h}) := \sum_{i\le N} e(\langle \mathbf{h}, x_i\rangle)$, $W(\mathbf{h}) := \prod_\nu (1+|h_\nu|)^{-1}$, $\mathcal{E} := \sum_{0<\|\mathbf{h}\|_\infty \le H} W(\mathbf{h})|\Sigma(\mathbf{h})|$.

A product expansion. Let $g_1, \dots, g_J$ be real trigonometric polynomials of degree $\le H$ with $|\hat{g}_\nu(h)| \le \gamma_\nu/(1+|h|)$ for all $|h| \le H$ and some $\gamma_\nu \ge 0$. Then $\prod_\nu g_\nu(y_\nu) = \sum_{\|\mathbf{h}\|_\infty \le H} \big(\prod_\nu \hat{g}_\nu(h_\nu)\big) e(\langle \mathbf{h}, \mathbf{y}\rangle)$, so

$$\Big|\sum_{i\le N} \prod_\nu g_\nu(x_{i,\nu}) - N \prod_\nu \hat{g}_\nu(0)\Big| \le \Big(\prod_\nu \gamma_\nu\Big)\,\mathcal{E}. \tag{25}$$

Upper bound. Apply this with $g_\nu = \chi_\nu^+$. By Proposition 31(iii) and, at $h = 0$, $\hat{\chi}_\nu^+(0) = |I_\nu| + O(\epsilon_H) \le 1 + \epsilon_H$, we may take $\gamma_\nu = 1 + \epsilon_H$, so $\prod \gamma_\nu \le (1 + \epsilon_H)^J \le e^{J\epsilon_H} \le e$. Also

$$\Big|\prod_\nu \hat{\chi}_\nu^+(0) - \mathrm{vol}(B)\Big| = \Big|\prod_\nu \big(|I_\nu| + \delta_\nu\big) - \prod_\nu |I_\nu|\Big| \le J\epsilon_H(1+\epsilon_H)^{J-1} \le e\, J\epsilon_H,$$

where $|\delta_\nu| \le \epsilon_H$, by the same telescoping as in Lemma 32 (each factor is at most $1 + \epsilon_H$). Since $\mathbf{1}_B \le \prod_\nu \chi_\nu^+$ by Proposition 31(i) and Lemma 32,

$$A \le \sum_i \prod_\nu \chi_\nu^+(x_{i,\nu}) \le N\,\mathrm{vol}(B) + e\,NJ\epsilon_H + e\,\mathcal{E}.$$

Lower bound. By Lemma 32 applied pointwise with $a_\nu = \mathbf{1}_{I_\nu}(y_\nu)$, $c_\nu = \chi_\nu^+(y_\nu)$, $b_\nu = \chi_\nu^-(y_\nu)$,

$$A \;\ge\; \sum_i \prod_\nu \chi_\nu^+(x_{i,\nu}) - \sum_{\nu=1}^J \; \sum_i \big(\chi_\nu^+ - \chi_\nu^-\big)(x_{i,\nu}) \prod_{\mu\ne\nu} \chi_\mu^+(x_{i,\mu}).$$

The first sum was bounded from below in the previous paragraph by $N\,\mathrm{vol}(B) - e\,NJ\epsilon_H - e\,\mathcal{E}$. For each ν, the inner sum is of the form treated by (25) with $g_\nu = \chi_\nu^+ - \chi_\nu^-$ and $g_\mu = \chi_\mu^+$ for $\mu \ne \nu$; here $|\hat{g}_\nu(h)| \le 2/(1+|h|)$ for $0 < |h| \le H$ by Proposition 31(iii), and $\hat{g}_\nu(0) = \hat{\chi}_\nu^+(0) - \hat{\chi}_\nu^-(0) \le 2\epsilon_H \le 2$, so $\gamma_\nu = 2$ and $\prod_\mu \gamma_\mu \le 2e$. Hence

$$\sum_i \big(\chi_\nu^+ - \chi_\nu^-\big)(x_{i,\nu}) \prod_{\mu\ne\nu} \chi_\mu^+(x_{i,\mu}) \le N \cdot 2\epsilon_H(1+\epsilon_H)^{J-1} + 2e\,\mathcal{E} \le 2e\,N\epsilon_H + 2e\,\mathcal{E}.$$

Summing over ν and combining, $A \ge N\,\mathrm{vol}(B) - 3e\,NJ\epsilon_H - 3e\,J\,\mathcal{E}$. Together with the upper bound this gives the theorem with $C = 3e$. □

Remark 34. *Two remarks on the two losses. (a) The factor $\log H$ inside ϵ_H comes from taking $s = \lceil \log H \rceil$ convolutions, which is what makes the truncation tail τ drop below $1/H$; with s bounded the tail is bounded below by an absolute constant and the construction fails. It is removed by Vaaler's polynomials, which we do not need. (b) The factor J on the exponential-sum term comes from the minorant, which is not non-negative and so is handled by Lemma 32 rather than by a product. In §7 both losses are absorbed: $J \ll (\log\log n)^{1/3}$ and $\log H \ll \log\log n$, while $H = (\log n)^2$, so $NJ\epsilon_H \ll N(\log n)^{-1}$ and the factor J is dominated by the saving of §B.8 with room to spare.*